Solutions to Magnetism Section of Numerical Examples in Physics

Solutions to Numerical Examples in Physics

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BHARATI BHAWAN (P & D)

Solutions to Numerical Examples in Physics

[For ISc Students of Indian Universities, Bihar Intermediate Education Council, Medical & Engineering Entrance Test, IIT, ISM (Dhanbad) and other competitive Examinations]

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IMPORTANT MATHEMATICAL RESULTS

1. Important trigonometric relations

(i) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

(ii) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

(iii) $\sin C + \sin D = 2\sin\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$

(iv) $\sin C - \sin D = 2\cos\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

(v) $\cos C + \cos D = 2\cos\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$

(vi) $\cos C - \cos D = -2\sin\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

(vii) $\sin 2A = 2\sin A \cos A$

(viii) $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

(ix) $\sin 3A = 3\sin A - 4\sin^3 A$

(x) $\cos 3A = 4\cos^3 A - 3\cos A$

2. Properties of a triangle

(i) sine property: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ where A, B and C are the angles and a, b and c are the sides opposite to A, B and C respectively.

(ii) cosine property: $c^2 = a^2 + b^2 - 2ab\cos C$; $a^2 = b^2 + c^2 - 2bc\cos A$ and $b^2 = c^2 + a^2 - 2ca\cos B$.

(iii) Area of a triangle $\Delta = \frac{1}{2}bc\sin A$ or $\frac{1}{2}ca\sin B$ or $\frac{1}{2}ab\sin C$.

(iv) In any small angled isosceles triangle: The small side = The small angle (in radian) × either of the longer sides.

3. Expansion formula

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1\cdot 2}x^2 + \frac{n(n-1)(n-2)}{1\cdot 2\cdot 3}x^3 + \ldots\ldots$$

When x is very small, $(1+x)^n = 1 + nx$.

$\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \ldots\ldots \quad (x < 1)$

$\ln(1-x) = -(x + x^2/2 + x^3/3 + x^4/4 + \ldots\ldots)$

$e^x = 1 + x/1! + x^2/2! + x^3/3! + x^4/4! + \ldots\ldots$

4. Important differentials

(i) $y = x^n$	$dy = nx^{n-1}dx$
(ii) $y = e^{ax}$	$dy = ae^{ax}dx$
(iii) $y = \sin ax$	$dy = a \cos axdx$
(iv) $y = \cos ax$	$dy = -a \sin axdx$
(v) $y = \tan ax$	$dy = a \sec^2 axdx$
(vi) $y = \cot ax$	$dy = -a \operatorname{cosec}^2 axdx$
(vii) $y = \sec ax$	$dy = a \sec ax \tan axdx$
(viii) $y = \operatorname{cosec} ax$	$dy = -a \operatorname{cosec} ax \cot axdx$

5. Important Integrals

(i) $\int x^n dx = \frac{x^{n+1}}{n+1}$ when $n \neq -1$ and when $n = -1 \int \frac{dx}{x} = \ln x$

(ii) $\int e^{ax}dx = \frac{e^{ax}}{a}$

(iii) $\int \sin axdx = -\frac{\cos ax}{a}$

(iv) $\int \cos axdx = \frac{\sin ax}{a}$

(v) $\int \tan axdx = -\frac{1}{a} \ln \cos x$

(vi) $\int \cot axdx = \frac{1}{a} \ln \sin ax$

(vii) $\int \sec^2 axdx = \frac{\tan ax}{a}$

(viii) $\int \operatorname{cosec}^2 axdx = -\frac{\cot ax}{a}$

(ix) $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$

(x) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$

(xi) $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln (x + \sqrt{x^2 \pm a^2})$

(xii) $\int \frac{dx}{\cos ax} = \frac{1}{a} \ln \tan\left(\frac{ax}{2} + \frac{\pi}{4}\right)$

(xiii) $\int \frac{dx}{\sin ax} = \frac{1}{a} \ln \tan \frac{ax}{2}$

(xiv) $\int \frac{dx}{x^2 - a^2} (x > a) = \frac{1}{2a} \ln \frac{x - a}{x + a}$

(xv) $\int \frac{dx}{a^2 - x^2} (a > x) = \frac{1}{2a} \ln \frac{a+x}{a-x}$

(xvi) $\int a^x dx = \frac{a^x}{\ln a}$

(xvii) $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a}$.

(xviii) $\int \sqrt{a^2 - x^2}\, dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$

(xix) $\int \sqrt{x^2 - a^2}\, dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln (x + \sqrt{x^2 - a^2})$

(xx) $\int \sqrt{a^2 + x^2}\, dx = \frac{x\sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \ln (x + \sqrt{x^2 + a^2})$

6. Maxima and Minima

The condition for maximum or minimum of a function y with respect to a variable x is that $\frac{dy}{dx} = 0$ and for maximum $\frac{d^2y}{dx^2}$ must be negative and for minimum $\frac{d^2y}{dx^2}$ must be positive. For illustration see Ex. 5 Chapter 4 and Ex. 7 Chapter 7.

7. Important Summations

(i) Sum of n natural numbers $= \frac{n(n+1)}{2}$.

(ii) Sum of squares of n natural numbers $= \frac{n(n+1)(2n+1)}{6}$.

(iii) Sum of the cubes of n natural numbers $= \left\{\frac{n(n+1)}{2}\right\}^2$.

(iv) Sum of an AP series $= \frac{1}{2} n$ (first term + last term).

(v) Sum of a GP series $= \frac{a(1 - r^n)}{1 - r}$, $r < 1$

$= \frac{a(r^n - 1)}{r - 1}$, $r > 1$.

Sum of an infinite GP series $= \frac{a}{1 - r}$.

(vi) Sum of infinite series

$$\sum_1^\infty \frac{1}{r^2} = \frac{\pi^2}{6} \cdot \sum_1^\infty \frac{1}{(2r+1)^2} = \frac{\pi^2}{8}.$$

$$\sum_{1}^{\infty}(-1)^{r+i}\frac{1}{r^2}=\frac{\pi^2}{12}\cdot\sum_{1}^{\infty}\frac{1}{(2r)^2}=\frac{\pi^2}{24}.$$

8. Most general method of solving simultaneous equations (e.g. three unknowns and three equations):

$$a_1x+b_1y+c_1z+d_1=0$$
$$a_2x+b_2y+c_2z+d_2=0$$
$$a_3x+b_3y+c_3z+d_3=0$$

is $$\frac{x}{x_{minor}}=\frac{-y}{y_{minor}}=\frac{z}{z_{minor}}=\frac{-1}{1_{minor}}$$

where minors are those of the elements of x, y, z and 1 of the determinant

$$\begin{vmatrix} x & y & z & 1 \\ a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{vmatrix}$$

9. Solid Angle: The solid angle subtended by an elementary area dS at a point distant r in direction θ from its normal is

$$d\omega=\frac{dS\cos\theta}{r^2}.$$

The solid angle of a cone of semivertical angle θ is:

$$\omega=2\pi(1-\cos\theta).$$

Solid angle in between directions θ and $\theta+d\theta=2\pi\sin\theta\,d\theta$.

10. Solution of two standard Differential Equations

$$\frac{d^2y}{dx^2}+\omega^2y=0 \qquad y=A\sin(\omega x+\alpha)$$

$$\frac{dy}{dx}+P(x)y=Q(x) \qquad y\,e^{\int P(x)dx}=\int e^{\int P(x)\,dx}\,Q(x)\,dx+C.$$

11. Radius of curvature of a curve

$$R=\frac{\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}}$$

□

Some Physical Constants

Speed of light	c	3×10^{8} m s^{-1}
Gravitational constant	G	6.67×10^{-11} N m^{2} kg^{-2}
Universal gas constant	R	8.31 J mol^{-1} K^{-1}
Permeability constant	μ_0	$4\pi \times 10^{-7}$ H m^{-1}
Permittivity constant	ε_0	8.85×10^{-12} F m^{-1}
Avogadro constant	N	6.02×10^{23} mol^{-1}
Boltzmann constant	k	1.38×10^{-23} J K^{-1}
Planck constant	h	6.63×10^{-34} J s
Elementary charge	e	1.60×10^{-19} C
Electron rest mass	m_e	9.11×10^{-31} kg
Electron magnetic moment	μ_e	9.27×10^{-24} J T^{-1}
Acceleration due to gravity (standard)	g	9.80 m s^{-2}
Standard atmospheric pressure	atm	1.01×10^{5} N m^{-2}
Density of water at 4°C		1000 kg m^{-3}
Rydberg constant	R	1.097×10^{7} m^{-1}
Loschmidt constant	n	2.69×10^{25} m^{-3}
Specific charge of electron	e/m	1.76×10^{11} C kg^{-1}
Proton rest mass	m_p	1.672×10^{-27} kg
Proton specific charge	e/m_p	9.59×10^{8} C kg^{-1}
Stefan constant	σ	5.67×10^{-8} W m^{-3} K^{-4}
Wien constant	b	2.9×10^{-3} m V
First Bohr orbit	r_0	5.29×10^{-11} m
Classical electron radius	r_e	2.82×10^{-15} m
Bohr magneton	μ_B	9.27×10^{-24} J T^{-1}
Nuclear magneton	μ_N	5.05×10^{-27} J T^{-1}
Proton magnetic moment	μ_p	2.792 μN
Neutron magnetic moment	μ_n	− 1.913 μN
Atomic mass unit	amu	1.66×10^{-27} kg or 931 MeV
	hc	1.989×10^{-25} J m
	$\frac{hc}{e}$	1.243×10^{-6} V m

□

Conversion Factors

Length:
$1\ m = 10^2\ cm = 39.37\ in$
$= 6.214 \times 10^{-4}\ mi$
$1\ mi = 5280\ ft = 1.609\ km$
$1\ in = 2.540\ cm$
$1\ Å\ (angstrom) = 10^{-8}\ cm$
$= 10^{-10}\ m$
$= 10^{-4}\ \mu\ (micron)$
$1\ \mu\ (micron) = 10^{-6}\ m$
$1\ AU\ (astronomical\ unit)$
$= 1.496 \times 10^{11}\ m$
$1\ light\ year = 9.46 \times 10^{15}\ m$
$1\ parsec = 3.084 \times 10^{16}\ m$

Angle:
$1\ radian = 57.3°$
$1° = 1.74 \times 10^{-2}\ rad$
$1' = 2.91 \times 10^{-4}\ rad$
$1'' = 4.85 \times 10^{-6}\ rad$

Area:
$1\ m^2 = 10^4\ cm^2 = 1.55 \times 10^{-5}\ in^2$
$= 10.76\ ft^2$
$1\ in^2 = 6.452\ cm^2$
$1\ ft^2 = 144\ in^2 = 9.29 \times 10^{-2}\ m^2$

Volume:
$1\ m^3 = 10^6\ cm^3 = 10^3\ litres$
$= 35.3\ ft^3 = 6.1 \times 10^4\ in^3$
$1\ ft^3 = 2.83 \times 10^{-2}\ m^3$
$= 28.32\ litres$
$1\ in^3 = 16.39\ cm^3$

Mass:
$1\ kg = 10^3\ g = 2.205\ lb$
$1\ lb = 453.6\ g = 0.4536\ kg$
$1\ amu = 1.6604 \times 10^{-27}\ kg$

Force:
$1\ N = 10^5\ dyn + 0.2248\ lbf$
$= 0.102\ kgf$
$1\ dyn = 10^{-5}\ N$
$= 2.248 \times 10^{-6}\ lbf$
$1\ lbf = 4.448\ N$
$= 4.448 \times 10^5\ dyn$
$1\ kgf = 9.81\ N$

Power:
$1\ W = 1.341 \times 10^{-3}\ hp$
$1\ hp = 745.7\ W$

Electric Charge:*
$1\ C = 3 \times 10^9\ stC$*
$1\ stC = \frac{1}{3} \times 10^{-9}\ C$

Current:*
$1\ A = 3 \times 10^9\ stA$
$1\ stA = \frac{1}{3} \times 10^{-9}\ A$
$1\ \mu A = 10^{-6}\ A,\ 1\ mA = 10^{-3}\ A$

Electric Field:*
$1\ N\ C^{-1} = 1\ V\ m^{-1}$
$= 10^{-2}\ V\ cm^{-1}$
$= \frac{1}{3} \times 10^{-4}\ stV\ cm^{-1}$

Electric Potential:*
$1\ V = \frac{1}{3} \times 10^{-2}\ stV$*
$1\ stV = 3 \times 10^2\ V$

Resistance:
$1\ \Omega = 10^6\ \mu\Omega$
$1\ M\Omega = 10^6\ \Omega$

Capacitance:*
$1\ F = 9 \times 10^{11}\ stF$*
$1\ stF = \frac{1}{9} \times 10^{-11}\ F$
$1\ \mu F = 10^{-6}\ F,\ 1\ pF = 10^{-12}\ F$

Magnetic Field:
$1\ T = 10^4\ gauss$
$1\ gauss = 10^{-4}\ T$

Magnetic Flux:
$1\ Wb = 10^8\ maxwell$
$1\ maxwell = 10^{-8}\ Wb$

Magnetizing Field:
$1\ A\ m^{-1} = 4\pi \times 10^{-3}\ oersted$
$1\ oersted = 1/4\pi \times 10^3\ A\ m^{-1}$

* In all cases 3 actually means 2.998 and 9 means 8.987; stC—stat coulomb (esu of charge); stV—stat volt (esu of voltage).

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❐

$$\therefore \quad 0 = \frac{\mu_0}{4\pi} \times \frac{2}{R-r} \times \frac{Ih}{2\pi R} - B$$

(The negative sign is used because the two fields are opposite)

$$\therefore \quad B = \frac{\mu_0 Ih}{4\pi R(R-r)}$$

10. *Two protons move parallel to each other with an equal velocity* $v = 3 \times 10^5$ m/s. *Find the ratio of forces of magnetic and electrical interaction of the protons.*

Solution.

$$F_{mag} = \frac{\mu_0}{4\pi} \frac{(qv)(qv)\sin 90°}{r^2} = \frac{\mu_0}{4\pi} \times \frac{q^2v^2}{r^2}$$

$$F_{elec} = \frac{1}{4\pi\varepsilon_0} \times \frac{q^2}{r^2}$$

$$\therefore \quad \frac{F_{mag}}{F_{elec}} = \mu_0\varepsilon_0 \times v^2$$

$$= 4\pi \times 10^{-7} \times 8.85 \times 10^{-12} \times 9 \times 10^{10} = 10^{-6}$$

EXERCISES
(A)

1. A current element of 2 milliampere metre is at the corner A of a unit cube ABCDEFGH, the element being along AB. Calculate the field at the remaining seven corners of the cube (figure 1.5).
2. Two current elements 2×10^{-4} A m and 5×10^{-4} Am are placed at the vertices *A* and *B* of an equilateral triangle along *AB* in the same sense. Calculate the magnetic field at *C*. (Each side of the triangle measures 10 cm)
3. Four current elements of length 0.01 cm and carrying 10^{-3} A lie at the four corners of a square *ABCD* along *AB*, *BC*, *CD* and *DA*. Calculate the magnetic field at the centre of the square. Each side of the square = 10 cm.
4. A current element of 0.2 A m is fixed at the centre of a circle of radius 10 cm. Another element of strength 0.5 A m is placed along the circumference and moved round the circle. At which positions will they exert maximum and minimum force on each other? Calculate these forces.
5. Two current elements of strengths 0.2 A m and 0.4 A m from a cross at the centre of a circle of radius 10 cm. Calculate the magnetic field at a point on the cricle 30° away (anticlockwise) from the first element.
6. Calculate the force experienced by an electron when it enters a uniform magnetic field of strength 2 T with velocity 0.2×10^8 m s^{-1} at right angles to the field. What kind of motion will it have in the magnetic field? (Charge of electron 1.6×10^{-19} coulomb.)

7. An electron of velocity 2×10^6 m s^{-1} is moving in a magnetic field of strength 1 T making an angle 45° with it. Find the magnitude and direction of the magnetic force experienced by it. Charge of electron = 1.6×10^{-19} C.

   [*Hint*: $\vec{F} = q_0 \vec{v} \times \vec{B}$.]

8. A ball of mass 2×10^{-3} kg carries a charge of 2×10^{-8} C. This ball is given an initial horizontal velocity of 5×10^4 m s^{-1} in the earth's gravitational field. What is the magnetic field required to keep the ball moving without deflection?

   [*Hint*: F_m (magnetic force) = $q_0 vB \sin\theta = \omega g$]

9. The plane of a rectangular coil of wire of dimensions 0.05 m × 0.10 m and carrying a current of 10 A, is parallel to a magnetic field of 0.1 T. What is the torque on the coil? What will be the torque when the normal to the coil makes an angle of 30° with the direction of the field?

10. Considering $\theta = \pi/2$ as the zero-energy position of a current loop of magnetic moment m placed in a magnetic field of strength B, find the minimum potential energy. In what position is the potential energy minimum? Is this an equilibrium position? What will be the work done in turning it through α from this position? Examine whether you get the same position for equilibrium and same value for work if $\theta = 0$ is taken as the zero–energy position.

(B)

11. A ring of dielectric material, with a total charge $+q$ distributed uniformly over its surface, rotates with angular speed ω. Find the magnetic moment of the ring. Compare its magnetic moment with its angular momentum, assuming that its mass is m.

12. A thin disc of dielectric material, with a total charge $+q$ distributed uniformly over its surface rotates with angular speed ω about an axis perpendicular to the disc and passing through its centre. Find the magnetic moment of the disc. The radius of the disc is r. Compare this magnetic moment with its angular momentum, assuming that its mass is m.

    [*Hint*: Consider a ring of radius x and thickness dx. Calculate charge on it and then the current constituted by its rotation. Find its magnetic moment and integrate for the whole disc.]

13. Electric charge $+q$ is uniformly distributed over the entire volume of a sphere of radius r. Calculate the magnetic moment of the sphere if it spins about its diameter with angular speed ω. Compare this moment with its angular momentum, assuming that its mass is m.

14. An insulated wire is wound so that it forms a flat coil with $N = 100$ turns. The radius of the innermost turn is $R_1 = 1$ cm, and of the outermost turn $R_2 = 4$ cm. What magnetic moment will this coil have when a current of $I = 10$ mA flows in it?

15. Apply Ampere's circuital law to calculate the magnetic field due to a toroid and a long straight solenoid in terms of linear density of turns n and current I.

16. Apply Ampere's circuital law to determine the magnetic induction inside a thick wire of radius R and carrying I amperes of current.

17. In a device called electromagnetic pump for transferring molten metals, a rectangular pipe is placed in a uniform magnetic field B perpendicular to its axis

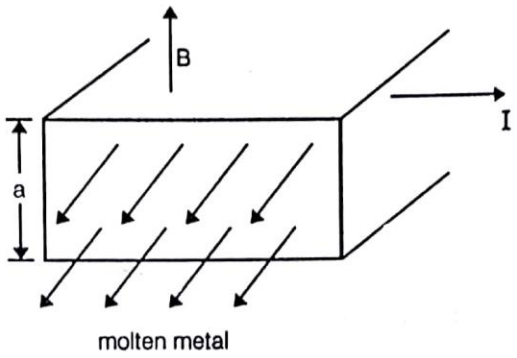


Fig. 1.10

and one of the sides of the section and a strong current I is then passed perpendicular to both B and the axis of the pipe. The molten metal is then passed through the pipe. What is the gauge pressure produced by the pump?

18. A current I flows through a long, thin-walled cylinder of radius R. What pressure do the walls of the cylinder experience?
19. A current I flows through a long, straight solenoid with radius R and linear density of turns n. Find the limiting current at which the winding may rupture if the tensile strength of the wire is σ.
20. What pressure does the surface of a long, straight solenoid with n turns per unit length experiences when a current I flows through it?
21. Find the magnitude and direction of the magnetic induction B due to an infinite plane carrying a current of linear density j. The density j is the same at all points of the plane.
22. A uniform current of surface density j flows inside an infinite plate of thickness $2d$ parallel to its surface. Find the magnetic induction inside and outside the plate with reference to distances from the middle plane of the plate. The magnetic permeability is assumed to be 1 both inside and outside.
23. A long horizontal wire AB, which is free to move in a vertical plane and carries a steady current of $I_1 = 20$ A, is in equilibrium at a height $h = 0.01$ m over another parallel wire CD, which is fixed in a horizontal plane and carries a steady current of $I_2 = 30$ A, as shown. Show that when AB is slightly depressed, it executes simple harmonic motion. Find the period of oscillations.

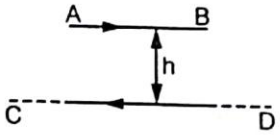


Fig. 1.11

[IIT 1994]

24. A wide copper strip of width l is bent into a piece of slender tubing of radius R with

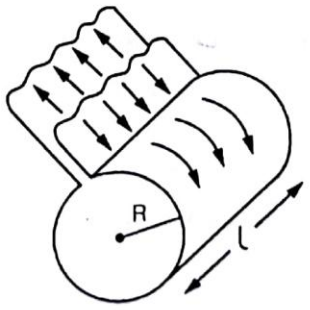


Fig. 1.12

two plane extensions as shown (Fig. 1.12). A current i flows through the strip, distributed uniformly over its width. In this way, one-turn solenoid has been formed (a) Derive an expression for the magnetic field in the tubular part.

25. Figure shows a cross-section of a long, thin ribbon of width l that is carrying a uniformly distributed stable current i into the page. Calculate the magnitude and the direction of the magnetic field B at a point P in the plane of the ribbon at a distance x from its edge.

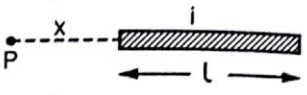


Fig. 1.13

26. In an 'electromagnetic rail gun' the projectile P is placed in between and in contact with two rails along which it can slide freely. A generator G provides a steady strong current that flows from one rail to another through the projectile. If a be the distance between rails, r is the radius of rails (presumed circular) and i the current, show that the *force* on the projectile is to the right and is given approximately by

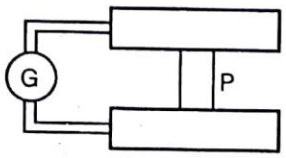


Fig. 1.14

$$F = \frac{1}{2}\frac{\mu_0 i^2}{R} \ln\left(1+\frac{a}{r}\right).$$

27. The current density inside a long solid, cylindrical wire of radius a is in the direction of the axis and varies linearly with linear distance r from the axis according to the law $j = j_0 \frac{r}{a}$. Find the magnetic field inside the wire. Express in terms of the total current i carried by the wire.

28. A long cylindrical conductor of radius a contains a long cylindrical hole of radius l. The axes of the two cylinders are parallel and are at a distance d apart. A current i is uniformly distributed over the cross-section. Find the magnetic field at the centre of the hole.

[*Hint:* Apply the principle of superposition of magnetic fields]

ANSWERS

1. At B, $B = 0$; at C, $B = 0.7 \times 10^{-10}$ T along $\vec{GC}$; at D, $B = 2 \times 10^{-10}$ T along $\vec{HD}$; at E, $B = 2 \times 10^{-10}$ T along $\vec{EH}$; at F, $B = 0.7 \times 10^{-10}$ T along $\vec{FG}$; at G, $B = 0.5443 \times 10^{-10}$ T perpendicular out of the plane ABG; at H, $B = 2 \times 10^{-10}$ T out of the plane ABH
2. 6.06×10^{-2} T.
3. 5.6×10^{-12} T along the normal to the square and out of the plane of it
4. 10^{-6} (maximum) when they are parallel and 0 (minimum) when they are perpendicular
5. 2.46×10^{-6} T, along the normal into the plane
6. 6.4×10^{-12} N; circular because magnetic force is always perpendicular to velocity
7. 2.26×10^{-13} N 8. 19.6 T 9. 5×10^{-3} N m; 2.5×10^{-3} N m
10. $-mB$ when $\theta = 0$, i.e., coil plane is perpendicular to B yes, **because** $\tau = 0$ in this position; $mB(1 - \cos\alpha)$

11. $\frac{1}{2}qwr^2,\ \frac{q}{2m}$ **12.** $\frac{1}{4}qwr^2,\ \frac{q}{2m}$ **13.** $\frac{1}{5}qwr^2,\ \frac{q}{2m}$

14. $m = \frac{1}{3}\pi NI\,(R_1^2 + R_1R_2 + R_2^2) = 2.2 \times 10^{-3}\ \text{A m}^2$

15. $B = \mu_0 nI;\ B = \mu_0 nI$ **16.** $B = \frac{\mu_0 Ir}{2\pi R^2}$ **17.** $p = \frac{IB}{a}$ **18.** $p = \frac{\mu_0 I^2}{8\pi^2 R^2}$

19. $I = \sqrt{\frac{2\sigma}{\mu_0 nR}}$ **20.** $p = \frac{1}{2}\mu_0 n^2 I^2$ **21.** $B = \frac{1}{2}\mu_0 j$

22. $B = \mu_0 jx$ (inside) and $B = \mu_0 jd$ (outside) **23.** $T = 2\pi\sqrt{\frac{h}{g}} = 0.2\ \text{s}$ **24.** $\frac{\mu_0 i}{l}$

25. $\frac{\mu_0 i}{2\pi l}\ln\left(1 + \frac{l}{x}\right)$ **27.** $\frac{\mu_0 i}{2\pi a^3}r^2$ **28.** $\frac{\mu_0 i\,d}{2\pi\,(a^2 - b^2)}$

CHAPTER 1

Introductory Notes

(A)

1. We have $\Delta B = \frac{\mu_0}{4\pi} \cdot \frac{I\Delta l \sin\theta}{r^2}$.

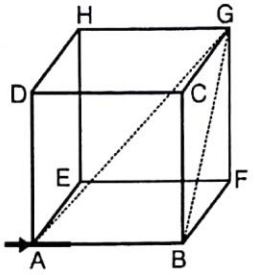


Fig. 1.1

At the corner B, $\Delta B = 0$ $(\because \theta = 0)$.

" " " C, $\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin 45°}{(\sqrt{2})^2}$

$= 0.7 \times 10^{-10}$ T.

" " " D, $\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin 90°}{(1)^2}$

$= 2 \times 10^{-10}$ T.

At the corner E, $\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin 90°}{(1)^2}$

$= 2 \times 10^{-10}$ T.

" " " F, $\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin 45°}{(\sqrt{2})^3}$

$(\because \quad AF = \sqrt{AB^2 + AF^2} = \sqrt{2})$

$= 0.7 \times 10^{-10}$ T.

" " " G, $\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin\theta}{AG^2}$.

Now $AG^2 = AB^2 + BG^2 = 1^2 + 1^2 + 1^2 = 3$

or $AG = \sqrt{3}$

$$\sin\theta = \frac{BG}{AG} = \frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{2}{3}} = 0.8165$$

$$\therefore \quad \Delta B \text{ at } G = 10^{-7} \frac{2 \times 10^{-3} \times 0.8165}{3} = 0.5443 \times 10^{-10} \text{ T.}$$

Also at corner H

$$\Delta B = 10^{-7} \times \frac{2 \times 10^{-3} \sin 90°}{(1)^2} = 2 \times 10^{-10} \text{ T.}$$

2. Field at C due to the element at A

$$= \frac{\mu_0}{4\pi} \times \frac{2 \times 10^{-4} \times \sin 60^\circ}{(0.01)^2}$$

$$= 10^{-7} \times \frac{2 \times 10^{-4} \times 0.866}{0.01} = 1.732 \times 10^{-9} \text{ T.}$$

This field is in the direction of a right hand screw perpendicular to AB and AC and rotating through 60° from $\vec{AB}$ to $\vec{AC}$.

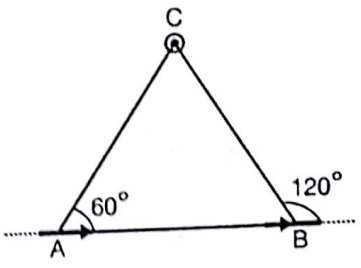


Fig. 1.2

This is a direction *perpendicular out of the plane of the paper.*

Field at C due to the element at B

$$= \frac{\mu_0}{4\pi} \times \frac{5 \times 10^{-4} \times \sin 120^\circ}{(0.1)^2} = 10^{-7} \times \frac{5 \times 10^{-4} \times \sin 120^\circ}{(0.1)^2}$$

$$= 4.320 \times 10^{-9} \text{ T.}$$

This field is in the direction of a right hand screw perpendicular to $\vec{AB}$ and $\vec{BC}$ and rotating through 120° from $\vec{AB}$ to $\vec{BC}$. This is also a direction perpendicular out of the plane of paper.

$\therefore$ Resultant field at $C = 6.06 \times 10^{-9}$ tesla.

3. The field at O due to the element at A

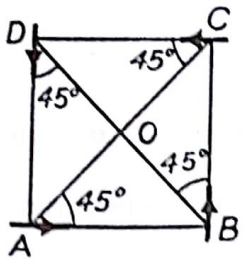


Fig. 1.3

$$= \frac{\mu_0}{4\pi} \times \frac{(10^{-3} \times 0.01 \times 10^{-2}) \times \sin 45^\circ}{(5\sqrt{2} \times 10^{-2})^2}$$

$$= 10^{-7} \times \frac{10^{-7} \times 0.707}{50 \times 10^{-4}} = 1.414 \times 10^{-12} \text{ T.}$$

Since all the elements are placed in the same sense and they are of the same strength, they produce the same magnetic field in the same direction at the centre of the square.

$\therefore$ Resultant field at $O = 4 \times 1.414 \times 10^{-12}$

$= 5.6 \times 10^{-12}$ T *perpendicular out of the plane of the paper.*

4. We have $\Delta F = \frac{\mu_0}{4\pi} \times \frac{(I\Delta l)(I'\Delta l')\sin 0}{r^2}$.

Therefore, force is maximum when $0 = \pi/2$ and minimum when $0 = 0$.

Thus $(\Delta F)_{max} = 10^{-7} \times \frac{0.2 \times 0.5}{(0.1)^2} = 10^{-6}$ newton,

$(\Delta F)_{min} = 0$.

5. $\Delta B_1 = 10^{-7} \times \frac{0.2 \sin 30°}{(0.01)^2} = 10^{-7} \times \frac{0.2 \times 0.5}{0.01} = 10^{-6}$ T,

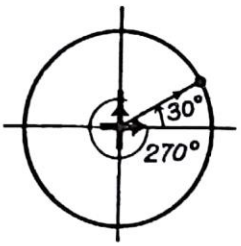


Fig. 1.4

perpendicular out of the plane of the paper.

$$\Delta B_2 = 10^{-7} \times \frac{0.4 \sin 60°}{(0.1)^2}$$

$$= 10^{-7} \times \frac{0.4 \times 0.866}{0.01}$$

$$= 3.464 \times 10^{-6} \text{ T},$$

perpendicular into the plane of the paper.

$\therefore$ Resultant field $= 3.464 \times 10^{-6} - 10^{-6}$

$= 2.461 \times 10^{-6}$ T,

perpendicular into the plane of the paper.

6. The force experienced by a charged body in a magnetic field is given by

$$\Delta \vec{F} = q_0 \vec{v} \times \vec{B} \quad \text{or} \quad \Delta F = q_0 vB \sin(\vec{v}, \vec{B}).$$

Here, $\sin(\vec{v}, \vec{B}) = \sin 90° = 1$

$\therefore$ $\Delta F = 1.6 \times 10^{-19} \times (0.2 \times 10^8) \times 2$

$= 6.4 \times 10^{-12}$ newton.

Motion is circular because force is always perpendicular to velocity.

7. $\vec{F} = q_0 \vec{v} \times \vec{B}$, i.e., $F = q_0 vB \sin\theta$ and it is in the direction of translation of a right hand screw perpendicular to $\vec{v}$ and $\vec{B}$ and rotating from $\vec{v}$ to $\vec{B}$ through θ.

$\therefore$ $F = 1.6 \times 10^{-19} \times 2 \times 10^6 \times 1 \times \sin 45°$

$= 3.2 \times 10^{-13} \times 0.707 = 2.26 \times 10^{-13}$ N.

8. When the gravitational pull is balanced by the magnetic force the ball will move undeflected.

i.e., F_m (magnetic force) $= q_0 vB \sin\theta = mg$.

Here $\theta = 90°$

$\therefore$ $B = \frac{mg}{q_0 v} = \frac{2 \times 10^{-3} \times 9.8}{2 \times 10^{-8} \times 5 \times 10^4}$

$= 1.96 \times 10 = 19.6$ tesla.

9. We have $\tau = ISB \sin\theta$

$= 10 \times (0.05 \times 0.1) \times 0.1 \sin 90° = 5 \times 10^{-3}$ N m.

When $\theta = 30°$,

$\tau = 10 \times (0.05 \times 0.1) \times 0.1 \sin 30° = 2.5 \times 10^{-3}$ N m.

10. We have $U_\theta = U_0 + mB(1 - \cos\theta)$ in general.

When $U = 0$ at $\theta = \frac{\pi}{2}$, $U_0 = -mB$.

$\therefore$ $U_\theta = -mB\cos\theta$.

Obviously U is minimum when $\theta = 0$

$U_{min} = -mB$.

Let us now take

$U = 0$ when $\theta = 0$

$0 = U_0 + mB(1 - \cos 0) = U_0$

$U = mB(1 - \cos\theta)$.

Work done in turning through $\theta = U - U_0$

$= -mB\cos\theta - (-mB) = mB(1 - \cos\theta)$ in the 1st case

$= mB(1 - \cos\theta) - 0 = mB(1 - \cos\theta)$ in the 2nd case.

Thus work done is the same whatever be the reference to zero-energy position.

(B)

11. Current $= \frac{q}{T} = \frac{q\omega}{2\pi}$

$\therefore$ μ (magnetic moment) $= \frac{q\omega}{2\pi} \cdot \pi r^2 = \frac{1}{2} q\omega r^2$

L (angular momentum) $= I\omega = mr^2\omega$

$\therefore$ $\frac{\mu}{L} = \frac{q}{2m}$.

12. Charge on elementary ring $= (2\pi x\, dx) \times \frac{q}{\pi r^2} = \frac{2q}{r^2} x\, dx$

$$\text{Current} = \frac{2qx\,dx}{r^2} \Big/ T = \frac{q\omega x\,dx}{\pi r^2}$$

$$\text{Magnetic moment} = \frac{q\omega x\,dx}{\pi r^2} \times \pi x^2 = \frac{q\omega}{r^2} x^3\,dx.$$

Denoting magnetic moment by μ,

$$\mu = \frac{q\omega}{r^2} \int_0^r x^3 dx = \frac{1}{4} q\omega r^2.$$

L (angular momentum) $= I\omega = \frac{1}{2} mr^2\omega$

$$\therefore \quad \frac{\mu}{L} = \frac{q}{2m}.$$

13. Consider a disc of thickness dx at a distance x.

Charge of the disc $= \pi(r^2 - x^2)\, dx \times \dfrac{q}{\frac{4\pi}{3} r^3} = \dfrac{3q}{4r^3} (r^2 - x^2)\, dx.$

Magnetic moment of the disc $= \frac{1}{4} \cdot \dfrac{3q}{4r^3} (r^2 - x^2)\, dx \times \omega \times (r^2 - x^2)$

(by the result of the solution of 12)

$$\therefore \quad \mu = \frac{2 \times 3q\omega}{16r^3} \int_0^r (r^2 - x^2)^2 x^2 dx = \frac{1}{5} q\omega r^2.$$

L (angular momentum) $= I\omega = \frac{2}{5} mr^2\omega$

$$\therefore \quad \frac{\mu}{L} = \frac{q}{2m}.$$

14. Consider a ring of radius x and thickness dx.

dm, magnetic moment of the ring = area × current $= \pi x^2 (ndx) I$

where n = number of turns per unit length $= \dfrac{N}{R_2 - R_1}$.

$$\therefore \quad m = n\pi I \int_{R_1}^{R_2} x^2 dx = \frac{1}{3} n\pi I (R_2^3 - R_1^3) = \frac{1}{3} \frac{N}{R_2 - R_1} \cdot \pi I (R_2^3 - R_1^3)$$

or $$m = \frac{1}{3} \pi NI (R_1^2 + R_1 R_2 + R_2^2).$$

Here $$m = \frac{1}{3} \pi \times 100 \times 10 \times 10^{-3} (1^2 + 1.4 + 4^2) \times 10^{-4}$$

$$= 2.2 \times 10^{-3} \text{ A m}^2.$$

15. Considering line integral of B along the loop $abcd$ in both figures

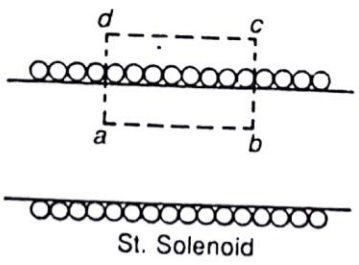


Fig. 1.5

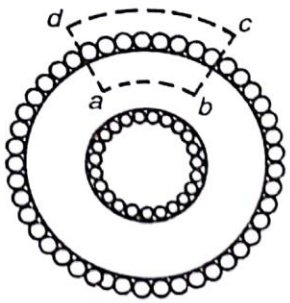


Fig. 1.6

we have

$$\int_{abcd} \vec{B} \cdot d\vec{l} = B\,ab + 0 + 0 + 0 = B \cdot ab = \mu_0 (nabI), \text{ by circuital law}$$

$$\therefore \qquad B = \mu_0 nI.$$

16. $$\int \vec{B} \cdot d\vec{l} = B \times 2\pi r = \mu_0 \frac{I}{\pi R^2} \times \pi r^2$$

or $$B = \frac{\mu_0 Ir}{2\pi R^2}.$$

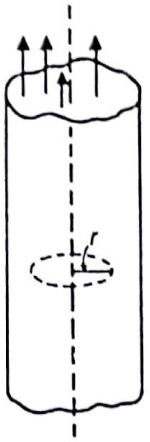


Fig. 1.7

17. $F_{mag} = (Il)\, B \sin 90°$ (from formula, $\Delta F = I\Delta lB \sin 0$)

$= IBa.$ $\quad \therefore \quad p = \frac{\text{force}}{\text{area}} = \frac{IBa}{a^2} = \frac{IB}{a}.$

18. Let us divide the cylinder into two parts: a thin strip dx and the remaining part. Then at a point P just outside,

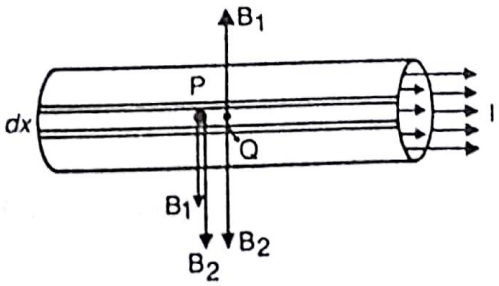


Fig. 1.8

$$B_1 + B_2 = \frac{\mu_0 I}{2\pi R},$$

but at a point Q inside, $B_1 + B_2 = 0$ because the field inside is zero.

$\therefore \quad B_1 = B_2$ (in magnitude)

$$\therefore \qquad B_2 = \frac{\mu_0 I}{4\pi R}.$$

Thus the strip may be supposed to be placed in magnetic field B_2.

$\therefore$ Force per unit length of the strip $= B_2 \left(\frac{Idx}{2\pi R} \right) \sin 90°$

(from formula, $\Delta F = I\Delta lB \sin 0$)

$$\therefore \quad p = \frac{\text{force}}{\text{area}} = \frac{B_2 I\,dx}{2\pi R\,dx} = B_2 \frac{I}{2\pi R} = \frac{\mu_0 I}{4\pi R} \cdot \frac{I}{2\pi R}$$

or $$p = \frac{\mu_0 I^2}{8\pi^2 R^2}.$$

19. Consider an element of length dx of the solenoid. We may consider it to be in a magentic field B_2 due to the remaining part of the solenoid.

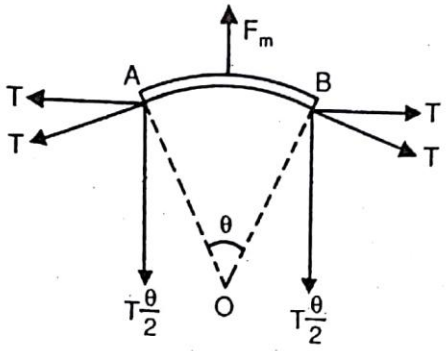


Fig. 1.9

$$B_2 = \frac{1}{2}\mu_0 nI \text{ (see solution of 18).}$$

Let T be the tension. Then inward force on the element

$$= T\frac{\theta}{2} + T\frac{\theta}{2} = T\theta.$$

Outward magnetic force

$$= B_2 Idx.$$

For equilibrium,

$$B_2 Idx = T\cdot\theta = T.\frac{dx}{R}.$$

$$\therefore \quad T = B_2 IR = \left(\frac{1}{2}\mu_0 nI\right) IR.$$

When $$T = \sigma, \quad \sigma = \frac{1}{2}\mu_0 nI^2 R$$

or $$I = \sqrt{\frac{2\sigma}{\mu_0 nR}}.$$

20. Let us consider an element of length dx. We may assume it to be placed in a uniform magnetic field B_2 due to the remaining part of the solenoid.

$$B_2 = \frac{1}{2}\mu_0 nI \text{ (see solution of 18)}$$

Force on it = $B_2\,(ndxI) = \frac{1}{2}(\mu_0 n^2 I^2 dx)$.

$$\therefore \quad p = \frac{\text{force}}{\text{area}} = \frac{1}{2}\mu_0 n^2 I^2.$$

21. Consider a closed path *abcd* and line integral of B along this path. The magnetic field at all points will be parallel to the plane, below to the right and above to the left.

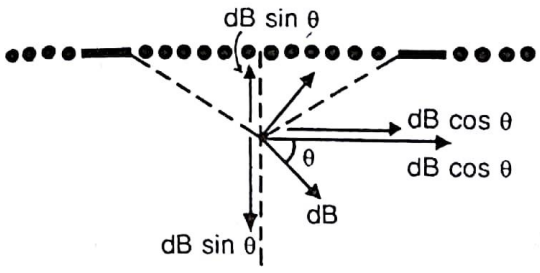


Fig. 1.10

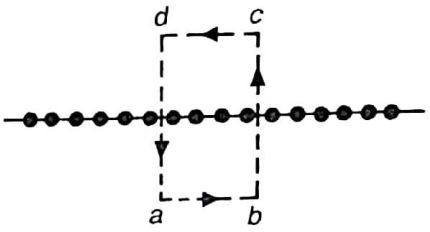


Fig. 1.11

$\therefore \quad \int_{abcd} \vec{B} \cdot d\vec{l} = Bab + 0 + Bab + 0 = \mu_0 (jab)$, by circuital law.

$\therefore \quad B = \frac{1}{2}\mu_0 j.$

22. $\int \vec{B} \cdot d\vec{l} = B \cdot b + 0 + B \cdot ab + 0 = \mu_0 (2xab\, j)$ or $B = \mu_0 jx.$

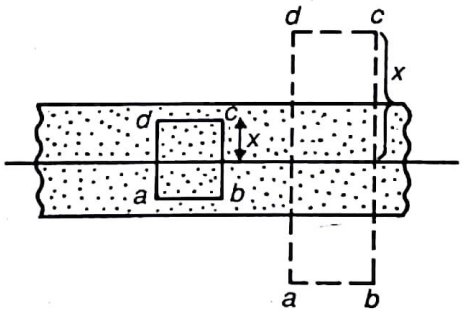


Fig. 1.12

$$\int \vec{B} \cdot d\vec{l} = B \cdot ab + 0 + B \cdot ab + 0 = \mu_0 (2d\, ab\, j)$$

or $B = \mu_0 jd.$

23. $F_m = \frac{\mu_0}{4\pi} \cdot \frac{(q_1 v_1)(q_2 v_2) \sin\theta}{d^2}$ and $F_e = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 q_2}{d^2}.$

Here $q_1 = q_2 = e, \quad \theta = 90°, \quad v_1 = v_2 = v.$

$\therefore \quad \frac{F_m}{F_e} = \frac{\mu_0}{4\pi} \times \frac{e^2 v^2}{d^2} \times \frac{4\pi\varepsilon_0 d^2}{e^2} = \mu_0 \varepsilon_0 v^2$

$\Rightarrow \frac{F_m}{F_e} = 4\pi \times 10^{-7} \times \frac{1}{36\pi \times 10^9} \times 10^{12} = \frac{1}{9 \times 10^4}.$

24. For equilibrium of Ab we have $F_m = F_{\text{gravity}}$

$$\frac{\mu_0}{2\pi} \times \frac{I_1 \cdot I_2}{h} \times l = (\mu l)\, g \quad \text{where } AB = l,\ \mu = \text{mass per unit length}$$

$$\Rightarrow \quad \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{h} = \mu g \qquad \ldots \text{(i)}$$

Let x be the instantaneous upward displacement. Then unbalanced upward force $= \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{h + x} \times l - \mu l g$

$$= \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{R + x} \times l - \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{h} \times l$$

$$= \frac{\mu_0}{2\pi} I_1 I_2 l \left[\frac{1}{h + x} - \frac{1}{h} \right]$$

$$= \frac{\mu_0}{2\pi} I_1 I_2 l \times \frac{-x}{(h + x)h}$$

$$= -\frac{\mu_0}{2\pi} \times \frac{I_1 I_2 x l}{h^2} \qquad (\because \quad x << h)$$

$$= (\mu l)\, \frac{d^2x}{dt^2} \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -\frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{\mu l h^2} \times x$$

Hence motion is simple harmonic.

$$\omega^2 = \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{\mu h^2}$$

Substituting the value of μ from (i)

$$\omega^2 = \frac{\mu g h}{\mu h^2} = \frac{g}{h}.$$

$$\Rightarrow \quad T = 2\pi \sqrt{\frac{h}{g}} = 2\pi \sqrt{\frac{0.01}{9.8}} = 0.2 \text{ s.}$$

25. By Ampere's circuital law

$$\left(\oint_{CP} \vec{B} \cdot \vec{dl} = \mu_0 i_{\text{inside}} \right)$$

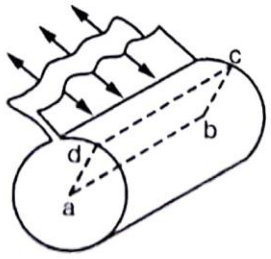


Fig. 1.12

Consider a loop $abcd$.

$$\int_{abcda} \vec{B} \cdot \vec{dl} = Bl$$

$$\therefore \quad Bl = \mu_0 i$$

$$\Rightarrow \quad B = \frac{\mu_0 i}{l}.$$

26. Consider an element dz of ribbon at a distance x from the edge. This element is a long straight conductor-carrying current $\frac{i}{l}dz$. Hence magnetic field at P is $\frac{\mu_0}{2\pi} \times \frac{\frac{i}{l}dz}{x+z}$ in the upward direction.

$$\therefore \qquad B = \frac{\mu_0 i}{2\pi l} \int_0^l \frac{dx}{x+z} = \frac{\mu_0 i}{2\pi l} \left[\ln (x+z)\right]_0^l$$

$$= \frac{\mu_0 i}{2\pi l} \left[\ln \frac{x+l}{x}\right] = \frac{\mu_0 i}{2\pi l} \ln \left(1 + \frac{l}{x}\right).$$

27. The two rails produce a strong magnetic field in the space between the rails and so the projectile experienced a strong force.

Consider an element dx of the projectile at a distance x from the axis of the upper rail.

$$B_1 \text{ (magnetic field at } dx \text{ due to the upper rail)} = \frac{\mu_0 i}{2\pi x}$$

$$B_2 \text{ (magnetic field at } dx \text{ due to the lower rail)} = \frac{\mu_0 i}{2\pi (2r + a - x)}$$

$$\therefore \qquad B = B_1 + B_2 = \frac{\mu_0 i}{2\pi x} + \frac{\mu_0 i}{2\pi (2r + a - x)}$$

$$= \frac{\mu_0 i}{2\pi} \left[\frac{1}{x} + \frac{1}{2r + a - x}\right] = \frac{\mu_0 i}{2\pi} \times \frac{2r + a}{x (2r + a - x)}$$

dF, force on the element $= Bidx \sin 90°$

$$= \frac{\mu_0 i^2}{2\pi} \times \frac{2r + a}{x (2r + a - x)} dx$$

$$\therefore \qquad F = \frac{\mu_0 i^2 (2r + a)}{2\pi} \int_r^{a+r} \frac{dx}{x (2r + a - x)}$$

Since x is small in comparison to $2r + a$

$$F = \frac{\mu_0 i^2 (2r + a)}{2\pi} \int_r^{a+r} \frac{dx}{x (2r + a)}$$

$$\Rightarrow \qquad F = \frac{\mu_0 i^2}{2\pi} \ln \left(1 + \frac{a}{r}\right)$$

28. Consider a concentric circular path.

$$\int_{CP} \vec{B} \cdot \vec{dl} = \int_{CP} B\,dl \cos 0^\circ = \int_{CP} B\,dl = B \cdot 2\pi r.$$

To find current inside consider a ring of radius x $(x < r)$ and width dx.

Current through ring $= \left(\frac{2\pi}{a} x\,dx\right) j = 2\pi x\,dx\, j_0 \frac{x}{a}$.

$$\therefore \quad i \text{ (total current)} = \int_0^a 2\pi x\,dx\; j_0 \frac{x}{a}$$

$$= \frac{2\pi j_0}{a} \times \frac{a^3}{3} = \frac{2\pi}{3} j_0 a^2$$

$$\text{Current inside } r = \int_0^r 2\pi x\,dx\, j_0 \frac{x}{a} = \frac{2\pi j_0}{a} \times \frac{r^3}{3}.$$

$$\therefore \quad B \cdot 2\pi r = \mu_0 \left(\frac{2\pi j_0}{a} \times \frac{r^3}{3}\right)$$

Putting the value of j_0 in forms of i

$$B \cdot 2\pi = \mu_0 \frac{2\pi}{a} \cdot \left(\frac{3I}{2\pi a^2}\right) \frac{r^2}{3}$$

$$\Rightarrow \quad B = \frac{\mu_0 i}{2\pi a^3} r^2.$$

29. $B = B_1 + B_2$ where B = total field due to the entire conductor, B_1 = field due to the cylindrical conductor filling the hollow space, B_2 = field due to the remainder. Draw a circular closed path.

By circuital law,

$$\int_{CP} \vec{B} \cdot \vec{dl} = \mu_0 i$$

$$B \cdot 2\pi d = \mu_0 \times \frac{i}{\pi (a^2 - b^2)} \times \pi d^2$$

$$\Rightarrow \quad B = \frac{\mu_0 id}{2\pi (a^2 - b^2)}$$

$B_1 = 0$ (because field at the centre of a cylindrical conductor is zero)

$$\therefore \quad B_2 = \frac{\mu_0 id}{2\pi (a^2 - b^2)}.$$

□□□

Chapter 2 – Magnetic Induction

Hence m is of magnitude 1.47×10^{-9} A m^2 against the field.

5. *The magnetic moment of a gadolinium atom is* 7.95 μ_B (μ_B = *Bohr magneton). Gadolinium crystallizes in a face-centred cubic lattice with the lattice constant* 3.2 Å. *Find the saturation magnetization.* $\mu_B = 9.27 \times 10^{-24}$ A m^2.

Solution.

$M_{sat} = n_0 m_{atom}$ where n_0 = no. of atoms per unit volume

An elementary cell of a face-centred lattice contains four atoms

So there are four atoms per unit cell i.e. in a volume a^3.

$$\therefore \quad n_0 = \frac{4}{a^3}$$

$$\therefore \quad M_{sat} = \frac{4}{a^3} \times m_{atom} = \frac{4}{3.2^3 \times 10^{-30}} \times 7.95 \times 9.27 \times 10^{-24}$$

$$= 9.0 \times 10^6 \text{ A m}^{-1}$$

EXERCISES
(A)

1. A magnet of magnetic moment 3 A m^2 weighs 75 g. If the density of the material of the magnet is 7900 kg m^{-3}, find its magnetization.

2. A ring iron of permeability 8.2×10^{-3} henry per metre has 500 windings on it. The mean circumference of the ring is 50 cm. It carries a current of 100 mA. Calculate, (i) the magnetic field H, (ii) the magnetic field B inside the core, (iii) the relative permeability of the core. (iv) By what factor is the magnetic field B due to the electronic loop currents in the core greater than that due to free current?

[*Hint:* (i) $H = nI$, (ii) $B = \mu H$, (iii) $\mu_r = \frac{\mu}{\mu_0}$, (iv) Required factor $= \frac{\mu_0 M}{\mu_0 H} = \chi_m$]

3. The dipole moment of an iron atom is 1.8×10^{-23} A m^2. If all the atoms of an iron rod of length 10 cm and diameter 1 cm are aligned along its length, what is the dipole moment of the bar? (Density of iron is 7870 kg m^{-3} and its atomic weight is 55.87 and Avogardo's number = 6.02×10^{26} kg mol^{-1})

4. A heavy steel beam of length 10 m and area of cross-section 0.25 m^2 runs parallel to the magnetic meridian. The horizontal component of the H-field of the earth is 30 A m^{-1}. A small compass needle is moved along the vertical line passing through the middle point of the beam and is found to turn through 180° at a distance of 5 m from it. Calculate the intensity of magnetization of the bar and also the susceptibility of its material.

5. A ring has a cross-section of 2 cm^2, a mean length of 30 cm and is wound with 400 turns. Find the current in the winding that is required to set up a magnetic field of 0.1 T in the ring. (Permeability of the material of the ring = 16×10^{-4} H m^{-1})

6. A solenoid of length 20 cm contains 150 turns. If a rod of mild steel of volume 2 cc and relative permeability 2000 is placed at the centre of the solenoid and 1 A

current is passed through the solenoid, what maximum magnetic moment will be acquired by the rod?

[*Hint:* $H = nI$; $\chi_m = \dfrac{M}{H}$; $\chi_m = \dfrac{M}{H}$; $\chi_m = \mu_r - 1$.]

(B)

7. A current flows uniformly along a solid rod of radius R. Show that the B-field inside this rod, at a distance r from the axis, is $B = \dfrac{\mu_0 I r}{2\pi R^2}$.

   [*Hint:* Apply Ampere's circuital law.]

8. A paramagnetic gas consists of atoms with a dipole moment of 1.5×10^{-23} J T^{-1}. The temperature of the gas is 27°C and its density (no. of atoms per unit volume) is 2.0×10^{26} m^{-3}. What is the maximum magnetization of the sample? Compare the thermal energy of each dipole with the magnetic potential energy in a magnetic field of 1 T and hence predict whether maximum magnetization can ever be attained.

9. A certain substance crystallizes in a closely-packed bcc structure. The lattice constant of the crystal is 2 Å and dipole moment is 2×10^{-23} J T^{-1}. Calculate the possible saturation magnetization of the substance.

   [*Hint:* There are 2 atoms per unit cell.]

10. A thin conducting strip of width h is wound in the shape of a solenoid of radius R. A steady current I flows through the strip. Find the magnetic induction inside and outside the solenoid as a function of the distance r from the axis.



Fig. 2.4

11. A wire is wound on a wooden toroidal core of very small cross-section with N turns. A current I flows through the wire. Find the ratio of the magnetic induction inside the core to that at the centre of the toroid.

    [*Hint:* For the centre, the toroidal conductor behaves like a circular conductor]

ANSWERS

1. 3.16×10 A m^{-1} **2.** (i) 100 A m^{-1}, (ii) 0.82 T, (iii) 6.5×10^3, (iv) 6.5×10^3

3. 12.0 A m^2 **4.** 5.3×10^4 A m^{-1}, 1767 **5.** 0.046 A **6.** 3 A m^2

8. 3.0×10^3 A m^{-1}, 414, No **9.** 5×10^6 A m^{-1}

10. $B = \dfrac{\mu_0 I}{h}\sqrt{1 - \left(\dfrac{h}{2\pi R}\right)^2}$ $r < R = \dfrac{\mu_0}{2\pi}\dfrac{I}{r}$ $r > R$ **11.** $\dfrac{N}{\pi}$.

CHAPTER 2

Magnetic Induction

1. Volume $= \dfrac{\text{mass}}{\text{density}} = \dfrac{75 \times 10^{-3}}{7900}\ \text{m}^3$.

M (magnetisation) $= \dfrac{m \text{ (magnetic moment)}}{V \text{ (volume)}}$.

$$\therefore \quad M = \frac{3}{\dfrac{75 \times 10^{-3}}{7900}} = \frac{3 \times 7900}{75 \times 10^{-3}} = 3{\cdot}16 \times 10^5\ \text{A m}^{-1}.$$

2. (i) $H = nI$

where n = number of turns per unit length, I = current in ampere.

$$\therefore \quad H = \left(\frac{500}{50 \times 10^{-2}}\right) \times 100 \times 10^{-3} = 100\ \text{A m}^{-1}.$$

(ii) $B = \mu H = 8{\cdot}2 \times 10^{-3} \times 100 = 0.82$ tesla.

(iii) $\mu = \mu_0 \mu_r$

$$\therefore \quad \mu_r = \frac{\mu}{\mu_0} = \frac{8{\cdot}2 \times 10^{-2}}{4\pi \times 10^{-7}} = 6.5 \times 10^3.$$

(iv) We have $B = \mu_0 (H + M) = \mu_0 H + \mu_0 M$.

$\mu_0 H$ is the field due to free current and $\mu_0 M$ is the field due to electronic loop currents.

$$\therefore \quad \text{The required factor} = \frac{\mu_0 M}{\mu_0 H} = \frac{M}{H} = \chi_m = \mu_r - 1$$

$$= 6{\cdot}5 \times 10^3 - 1 \approx 6.5 \times 10^3.$$

3. Mass of the rod $= \pi\,(0.005)^2 \times 0.1 \times 7870$ kg

$$\text{No. of atoms in the rod} = \frac{6.02 \times 10^{-6}}{55.87} \times \pi\,(0.005)^2 \times 0.1 \times 7870$$

$$= 6.66 \times 10^{23}$$

$$\therefore \text{ Dipolemoment of the rod.} = 6.66 \times 10^{23} \times 1.8 \times 10^{-23}\ \text{Am}^2$$

$$= 11.988 = 12\ \text{Am}^2$$

4. At the neutral point the needle turns through 180°. We have at the neutral point,

$$H = \frac{m}{4\pi (d^2 + l^2)^{3/2}}$$

$$\therefore \qquad 30 = \frac{m}{4\pi(5^2+5^2)^{3/2}} \qquad \text{or} \qquad m = 30 \times 250\sqrt{2} \times 4\pi$$

$$M = \frac{m}{V} = \frac{30 \times 250\sqrt{2} \times 4\pi}{10 \times 0.25} = 5.3 \times 10^4 \text{ A m}^{-1}$$

$$\chi_m = \frac{M}{H} = \frac{5.3 \times 10^4}{30} = 1767.$$

5. We have $B = \mu H = \mu nI \qquad \therefore \quad 0.1 = 16 \times 10^4 \times \frac{400}{30 \times 10^{-2}} \cdot I$

$$I = \frac{0.1 \times 30 \times 10^{-2}}{64 \times 10^{-2}} = 0.046 \text{ A}.$$

6. We have, $H = nI = \frac{150}{20 \times 10^{-2}} \times 1 = 750 \text{ A m}^{-1}$

$$\chi_m = \frac{M}{H}, \quad \text{or} \quad M = (\mu_r - 1)\, H \qquad (\because \ \chi_m = \mu_r - 1)$$

$$= (2000 - 1) \times 750 = 15 \times 10^5.$$

Now, $M = \frac{m}{V}$

or $m = 15 \times 10^5 \times 2 \times 10^{-6} = 3 \text{ A m}^2.$

7. The line integral along a concentric circular path of radius r

$$= B \times 2\pi r.$$

Current enclosed $= \frac{I}{\pi R^2} \times \pi r^2.$

By circuital law, $B \cdot 2\pi r = \mu_0 \frac{Ir^2}{R^2}$

or $B = \frac{\mu_0 Ir}{2\pi R^2}.$

8. $M_{max} = N \times m = 2.0 \times 10^{26} \times 1.5 \times 10^{-23} = 3.0 \times 10^3 \text{ A m}^{-1}$

$$E_{ther} = \frac{3}{2} kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300$$

$$E_{mag} = mB = 1.5 \times 10^{-23} \times 1$$

$$\therefore \qquad \frac{E_{ther}}{E_{mag}} = \frac{3/2 \times 1.38 \times 300}{1.5} = 414.$$

Hence, maximum magnetisation cannot be attained by the sample.

9. $M = Nm = \frac{2m}{a^3} = \frac{2 \times 2 \times 10^{-23}}{2^3 \times 10^{-30}} = 5 \times 10^6 \text{ Am}^{-1}.$

10. Inside the solenoid the magnetic field is parallel to the axis. Draw a line on the surface parallel to the axis and then flatten the strip. Now on the strip we will have parallelograms of shorter diagonal $2\pi R$ inclined to the edge of the strip by θ. $\sin\theta = \frac{h}{2\pi R}$. The sides of the parallelogram are, say, l. Then $h = l\cos\theta$.



Fig. 2..1

By Ampere's circuit law,

$$\oint_{abcd} \vec{B}\cdot\vec{dl} = Bl = \mu_0 I$$

$$\Rightarrow \qquad B = \frac{\mu_0 I}{l} = \frac{\mu_0 I}{h}\cos\theta = \frac{\mu_0 I}{h}\sqrt{1-\left(\frac{h}{2\pi R}\right)^2}.$$

For outside points : The conductor produces field perpendicular to the plane passing through the axis and the field point because it behaves like a straight conductor. So we consider a circular path of radius r in a plane perpendicular to the axis

$$\oint \vec{B}\,.\,\vec{dl} = B\,2\pi r.$$

By circuital law, $B\cdot 2\pi r = \mu_0 I$

$$B = \frac{\mu_0}{2\pi}\cdot\frac{I}{r}.$$

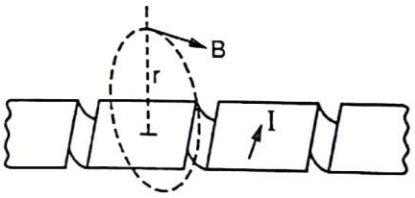


Fig. 2.2

11. Let R be the mean radius of the toroid. Inside the core the field is *along* the axis. So consider a closed circular path coinciding with the axis.

$$\oint \vec{B}\cdot\vec{dl} = B\cdot 2\pi r = \mu_0 (NI) \quad\Rightarrow\quad B = \frac{\mu_0 NI}{2\pi R}.$$

At the centre the field is perpendicular to the plane.

$$B_{centre} = \frac{\mu_0 I}{2R}.$$

$$\therefore \qquad \frac{B}{B_{centre}} = \frac{\mu_0 NI}{2\pi R}\times\frac{2R}{\mu_0 I} = \frac{N}{\pi}.$$

□ □ □

Chapter 3 – Magnetic Field and Potential

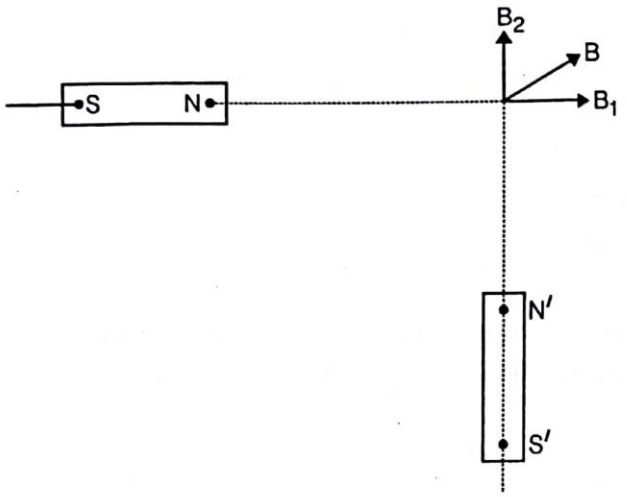


Fig. 3.4

Let it make an angle θ with B_1. Then $\tan\theta = \dfrac{B_2}{B_1} = \dfrac{1.56}{1.48}$

or $\theta = 45°51'$

EXERCISES
(A)

1. An electron is moving with a velocity of 2×10^6 m s^{-1} in the vicinity of a 0.1 m-long bar magnet of pole strength 50 A m in a direction perpendicular to the axis of the magnet at a distance of 0.15 m from the nearer pole. What is the force on the electron? Charge of electron = 1.6×10^{-19} C

   [*Hint:* Calculate B-vector of the field and apply $F = q_0 v B \sin\theta$.]

2. A magnetic needle of magnetic moment 0.9 A m^2 and pole strength 5 A m is pivoted so that it is free to turn in a horizontal plane at a position where the earth's B-field is 3.6×10^{-5} tesla. It is in equilibrium at an angle 30° from the magnetic meridian when it is pulled by a string attached to its north pole in the easterly direction. What is the tension of the string?

3. A magnet of length 0.1 m and pole strength 0.1 A m is freely suspended in a horizontal uniform magnetic field of intensity 0.18×10^{-4} T. Find the torque tending to restore the magnet to its original position of rest when it is deflected by 30° from that position.

4. A bar magnet lies in a magnetic field of intesity 0.5×10^{-4} tesla. The torque required to deflect it through 30° is 2.5×10^{-5} N m. Calculate the magnetic moment of the magnet.

5. A magnet of length 0.2 m balances the earth's horizontal field at a point 0.2 m from the nearer pole. The magnet is placed with its north pole pointing south. Find its magnetic moment. [Horizontal induction of earth's field = 3.6×10^{-5} tesla.]

6. The intensities of magnetic field at two points on the axis of a bar magnet at distances 0.1 m and 0.2 m from its middle point are in the ratio 12.5 : 1. Calculate the distance between the poles of the magnet.

7. Two magnets of moments m and 2 m are mounted to form a cross. The combination rests horizontally when suspended by a silk fibre. Find the angle which the weaker magnet makes with the meridian.
8. A short magnet is placed horizontally in the magnetic meridian with its north pole pointing north. It is then found that there is a neutral point 20 cm from the middle of the magnet. If the magnet is turned through 180°, where will the neutral point be?

   [*Hint:* When magnet is placed with its north pole pointing north, neutral point occurs at the broadside-on position. When it is rotated through 180°, the neutral point will shift to the end-on position.]
9. The intensities of a magnetic field at two points on the axis of a bar magnet at distances of 0.15 m and 0.2 m from its middle point are in the ratio 75 : 20. Find the distance between the poles of the magnet.
10. A short bar magnet is placed horizontally in the magnetic meridian with its north pole pointing north and the neutral points are found 0.15 m from the centre of the magnet. Find the distance between the neutral points if the magnet is reversed.
11. A neutral point is found on the prolongation of the axis of a bar magnet at a distance of 0.1 m from the nearest pole. If the length of the bar be 0.1 m and the B-field of earth $= 0.36 \times 10^{-4}$ tesla, find the pole strength of the magnet.
12. A magnet whose poles are 12 cm apart is placed in the magnetic meridian. The field due to this magnet counterbalances the earth's horizontal field ($H = 30$ A m^{-1}) at a point 10 cm from each pole. Find the pole strength of the magnet.
13. A uniformly magnetized bar of brittle steel is broken into two pieces, one twice as long as the other, and the pieces are fastened together at right angles to each other. How would the combination thus formed set itself under the action of the earth's magnetic force, if made to float on water?
14. Calculate the moment of a magnet, which when placed at right angles to the horizontal magnetic field of the earth of intensity 30 A m^{-1} experiences a couple of 3×10^{-4} newton metre.

    [*Hint:* $B_0 = \mu_0 H_0 = 4\pi \times 10^{-7} \times 30$ tesla and $\tau = mB_0 \sin\theta$]
15. A neutral point is found on the prolongation of the axis of a bar magnet of length 10 cm at 10 cm from the nearer pole. How has the magnet been placed with respect to the magnetic meridian? Calculate the magnetic moment of the magnet given that B-vector of earth's field along the horizontal $= 3.6 \times 10^{-5}$ tesla. Also calculate the torque required to deflect it through 30° from the magnetic meridian.

(B)

16. Two bar magnets of length 0.1 m and pole strength 75 A m each, are placed on the same line. The distance between their centres is 0.2 m. What is the resultant force due to one on the other when (i) the north pole of one faces the south pole of the other, (ii) the north pole of one faces the north pole of the other?
17. A bar magnet whose poles are 10 cm apart is placed in the magnetic meridian with its north pole pointing south. The neutral point is at a distance of 20 cm from the nearest pole. Find the intensity of the resultant field at a point on the perpendicular bisector of the axis of the magnet and at a distance of 10 cm from the centre of the magnet. (B-vector of earth's field along the horizontal $= 4 \times 10^{-5}$ tesla)

18. A long, vertical magnet is placed at a perpendicular distance of 0.8 m from the centre of a horizontal magnetic needle of length 0.12 m and pole strength 0.6 A m. If the moment of the couple acting upon the needle is 72×10^{-7} N m, find the ploe strength of the long magnet.

[*Hint:* Since the magnet is long, the effect of the distant pole may be neglected.]

19. A magnet placed in the magnetic meridian with its north pole pointing north of the earth produces a neutral point at a distance 0.15 m from either pole. It is then broken into two equal pieces and one such piece is placed in a similar position. Calculate the position of the neutral point. Where will the neutral point be if the other piece is placed over the previous one with like poles together?

20. Two short magnets of the same type, but of moments m and $2m$ are mounted on a frame so as to form a cross. If the combination is suspended at the centre with a vertical fibre, find the direction in which it will set in the earth's magnetic field. Calculate also the magnetic B-field at a distance d from the centre of the cross on the prolongation of one of the arms.

21. Assuming $\theta = \pi/2$ as the zero-energy position of a magnet in a magnetic field, show that the potential energy of a magnet can be written as $U = -\vec{B} \cdot \vec{m}$. Hence find the stable equilibrium position of a magnet in a magnetic field.

22. Show that the force of attraction between two short magnets of moment m and m' with their centres d distance apart and their axes in the same direction along the same line is $\frac{\mu_0}{4\pi} \cdot \frac{6mm'}{d^4}$.

23. A bar magnet of moment $m = 4.0$ A m^2 lies on a table horizontally in the earth's magnetic field $H = 30$ A m^{-1}. A student turns it and holds it erect on the table. What is the work done by the student.

24. Find the force between two small circular coils of radii $r_1 = 1$ cm, $r_2 = 2$ cm and currents $I_1 = 1$ A and $I_2 = 2$ A placed at $d = 20$ cm apart with their planes parallel to each other.

ANSWERS

1. 4.6×10^{-17} N 2. 2.08×10^{-4} N 3. 9×10^{-8} Nm 4. 1 A m^2 5. 3.84 A m^2
6. 0.1 m 7. 63°26′ 8. 25.2 cm 9. 0.180 m 10. 0.378 m 11. 4.8 A m
12. 3.142 A m 13. 26°36′ (longer one) 14. 7.958 A m^2
15. 0.48 A m^2, 8.64×10^{-6} Nm
16. (i) 3.4×10^{-2} N (attraction) (ii) 3.4×10^{-2} N (repulsion)
17. 2.88 A m^2, 2.46×10^{-4} T 18. 645 A m
19. 0.119 m from either pole, 0.15 m from either pole of the combination
20. 26°36′, $\frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2}m}{d^3}$, $\frac{\mu_0}{4\pi} \cdot \frac{m\sqrt{17}}{d^3}$ 23. $\mu_0 mH = 1.5 \times 10^{-7}$ J
24. $\frac{\mu_0}{4\pi} \cdot \frac{6\pi^2 r_1^2 r_2^2 I_1 I_2}{d^4} = 2.3 \times 10^{-8}$ N

CHAPTER 3

Magnetic Field and Potential

(A)

1. $B = \frac{\mu_0}{4\pi} \cdot \frac{2md}{(d^2 - l^2)^2} = 10^{-7} \times \frac{2 \times (50 \times 0.1) \times 0.2}{(0.2^2 - 0.05^2)^2} = 1.4222 \times 10^{-4}\ T$

$\Delta F = q_0 v B \sin\theta = 1.6 \times 10^{-19} \times 2 \times 10^6 \times 1.422 \times 10^{-4} \sin 90°$

$= 4.6 \times 10^{-7}$ N.

2. τ, torque (magnetic) $= 2q_m Bl \sin\theta$.

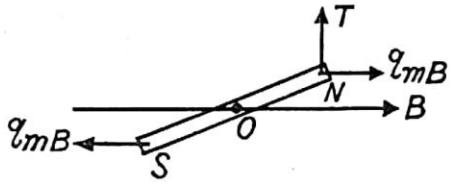


Fig. 3.1

Torque (mechanical) $= Tl \cos\theta$.

$\therefore \quad Tl \cos\theta = 2q_m Bl \sin\theta$

or $\quad T = 2q_m B \tan\theta$

$= 2 \times 5 \times 3.6 \times 10^{-5} \times \tan 30°$

$= 36 \times 10^{-5} \times 0.577$

$= 2.08 \times 10^{-4}$ N.

3. Torque $(\tau) = mB_0 \sin\theta = (0.1 \times 0.1) \times 0.18 \times 10^{-4} \times \sin 30°$

$= 0.01 \times 0.18 \times 10^{-4} \times 0.5 = 9 \times 10^{-8}$ N m.

4. $\tau = mB \sin\theta$

$2.5 \times 10^{-5} = m \times 0.5 \times 10^{-4} \sin 30°$

or $\quad m = \frac{2.5 \times 10^{-5}}{0.5 \times 10^{-4} \times 0.5} = 1$ A m^2 or J T^{-1}.

5. When a bar magnet is placed with its *north pole pointing south*, the neutral point occurs on the prolongation of the axis of the magnet.

$\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{2md}{(d^2 - l^2)^2} = B_0$

where B_0 = horizontal component of B-vector of the earth's magnetic field.

$\therefore \quad 10^{-7} \times \frac{2m \times 0.3}{(0.3^2 - 0.1^2)^2} = 3.6 \times 10^{-5}$

or $\quad m = \frac{0.0064 \times 3.6 \times 10^{-5}}{0.6 \times 10^{-7}} = 3.84$ A m^2.

6. $B_1 = \frac{\mu_0}{4\pi} \times \frac{2m \times 0.1}{(0.1^2 - l^2)^2}$; $B_2 = \frac{\mu_0}{4\pi} \times \frac{2m \times 0.2}{(0.2^2 - l^2)^2}$.

$\therefore \quad \frac{B_1}{B_2} = \frac{0.1}{0.2} \cdot \frac{(0.04 - l^2)^2}{(0.01 - l^2)^2}$, or $\frac{12.5}{1} = \frac{1}{2}\left(\frac{0.04 - l^2}{0.01 - l^2}\right)^2$

$\therefore \quad 2l = 0.1$ m.

7. Let the weaker magnet make an angle θ with the magnetic field.

τ_1 (magnetic torque on m) $= mB \sin\theta$ (anticlockwise)

τ_2 (magnetic torque on $2m$) $= 2mB \sin\left(\frac{\pi}{2} - \theta\right)$ (clockwise).

For equilibrium, $mB \sin\theta = 2mB \cos\theta$

or $\tan\theta = 2$ or $\theta = \tan^{-1} 2 = 63°26'$.

8. When a magnet is placed with its north pointing north, the neutral point is on the perpendicular bisector of the magnet.

$\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{m}{d^3} = B_0$. Here $d = 0.20$ m. $\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{m}{0.2^3} = B_0$ ... (i)

When the magnet is turned through 180°, its position becomes *north pointing south* when the neutral point is formed on the end. on position.

$\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3} = B_0$. ... (ii)

From (i) and (ii),

$$\frac{\mu_0}{4\pi} \cdot \frac{m}{0.2^3} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3}$$

or $d^3 = 0.2^3 \times 2$ or $d = 0.2 \times 2^{\frac{1}{3}} = 0.252$ m $= 25.2$ cm.

9. $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2m \times 0.15}{(0.15^2 - l^2)^2}$; $B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2m \times 0.2^2}{(0.2^2 - l^2)^2}$

$\therefore \quad \frac{B_1}{B_2} = \frac{0.15}{0.20}\left(\frac{0.2^2 - l^2}{0.15^2 - l^2}\right)^2 = \frac{75}{20}$.

Hence $2l = 0.180$ m.

10. We have $\frac{\mu_0}{4\pi} \cdot \frac{m}{0.15^3} = B_0$, and $\frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3} = B_0$

$\therefore \quad d^3 = 2 \times 0.15^3$ or $d = 0.15 \times 2^{\frac{1}{3}}$ m.

$\therefore$ The distance between neutral points

$$= 2d = 0.15 \times 2^{\frac{4}{3}} = 0.378 \text{ m}.$$

11. $\frac{\mu_0}{4\pi} \cdot \frac{2m \times 0.15}{(0.15^2 - 0.05^2)^2} = 0.36 \times 10^{-4}$ or $m = \frac{0.36 \times 10^{-4}}{0.30} \times 0.02^2 \times 10^7$

$$q_m = \frac{m}{2l} = \frac{0.36 \times 4 \times 10^{-4} \times 10^3}{0.3 \times 0.1} = 4.8 \text{ A m.}$$

12. Since the neutral point is 10 cm away from each pole, it is on the perpendicular bisector of the magnet.

$\therefore$ $\frac{\mu_0}{4\pi} \cdot \frac{m}{(d^2 + l^2)^{3/2}} = B_0 = \mu_0 H_0$ or $\frac{m}{(d^2 + l^2)^{3/2}} = 4\pi H_0$

or $\frac{m}{x^3} = 4\pi H_0$ where $x = \sqrt{d^2 + l^2}$ or $m = 4\pi \times 30 \times 0.1^3$

and $q_m = \frac{4\pi \times 30 \times 0.1^3}{0.12} = 3.142 \text{ A m.}$

13. On breaking a magnet pole-strength remains unchanged. Hence the magnetic moment of the longer part will be twice the magnetic moment of the shorter one. Let the longer one make an angle θ with the magnetic meridian.

$$\tau_1 = 2mB \sin\theta; \quad \tau_2 = mB \sin\left(\frac{\pi}{2} - \theta\right).$$

For equilibrium,

$$\tau_1 + \tau_2 = 0 \quad \text{or} \quad \tau_1 = -\tau_2$$

$\therefore$ $\tau_1 = \tau_2$ (numerically)

or $2mB \sin\theta = mB \cos\theta$

or $\tan\theta = 0.5$ or $\theta = \tan^{-1} 0.5 = 26°36'$.

14. We have $\tau = B_0 m \sin\theta = \mu_0 H_0 m \sin\theta$ ($\because$ $B_0 = \mu_0 H_0$)

or $3 \times 10^{-4} = 4\pi \times 10^{-7} \times 30 \times m \sin 90°$

or $m = 7.958 \text{ A m}^2$.

15. The magnet is placed with *north pole pointing south of the earth.*
We have at the neutral point on the end-on position,

$$B_0 = \frac{\mu_0}{4\pi} \cdot \frac{2md}{(d^2 - l^2)^2},$$

or $3.6 \times 10^{-5} = 10^{-7} \cdot \frac{2m \times 0.15}{(0.15^2 - 0.05^2)^2}$

or $m = \frac{3.6 \times 10^{-5} \times 0.0004}{0.30 \times 10^{-7}} = 0.48 \text{ A m}^2.$

We have $\tau = mB \sin\theta = 0.48 \times 3.6 \times 10^{-5} \sin 30° = 8.64 \times 10^{-6} \text{ N m.}$

(B)

16. $F_{N,S'} = \frac{\mu_0}{4\pi} \cdot \frac{q_m q'_m}{r^2} = 10^{-7} \cdot \frac{75 \times 75}{(0.1)^2} = 5.625 \times 10^{-2}$ N (attraction)

$F_{N',S} = 10^{7} \times \frac{75 \times 75}{(0.3)^2}$

$= 0.625 \times 10^{-2}$ N (attraction)

$F_{NN'} = F_{SS'} = 10^{-7} \times \frac{7 \times 75}{0.2^2}$

$= 1.406 \times 10^{-2}$ N (repulsion)

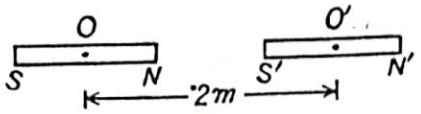


Fig. 3.2

(i) $\therefore$ Resultant force of attraction $= F_{N,S'} + N_{N'}S - 2F_{NN'}$

$= (5.625 + 0.625 - 2 \times 1.406) \times 10^{-2}$ N

$= 3.4 \times 10^{-2}$ N (attraction).

(ii) There will be force of repulsion of the same magnitude.

17. When a magnet is placed with its north pole pointing south, the neutral point is on the prolongation of the axis of the magnet.

$\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{2md}{(d^2 - l^2)^2} = B_0 \quad$ or $\quad m = \frac{4\pi}{\mu_0} \cdot \frac{B_0 (d^2 - l^2)^2}{2d}$

or $\quad m = 10^{7} \times \frac{4 \times 10^{-5} (0.25^2 - 0.05^2)^2}{2 \times 0.25}$

$= \frac{4 \times 0.0036}{0.5} \times 10^2 = \frac{144}{50}$ A m^2 $= 2.88$ A m^2.

Intensity of magnetic field at a point on the perpendicular bisector is

$$B = \frac{\mu_0}{4\pi} \cdot \frac{m}{(d^2 + l^2)^{3/2}} = 10^{-7} \times \frac{\frac{144}{50}}{(0.1^2 + 0.05^2)^{3/2}}$$

$$= \frac{144}{50 \times (0.0125)^{3/2}} \times 10^{-7} = \frac{2.88}{1.397 \times 10^{-3}} \times 10^{-7}$$

$$= 2.06 \times 10^{-4} \text{ tesla.}$$

Earth's horizontal intensity at that point is B_0 in the direction of B.

$\therefore$ The resultant intensity at that point

$= B + B_0 = (2.06 + 0.4) \times 10^{-4} = 2.46 \times 10^{-4}$ tesla.

18. Let q_m be the pole-strength of the long magnet. Since the magnet is long, the effect of the distant pole may be neglected.

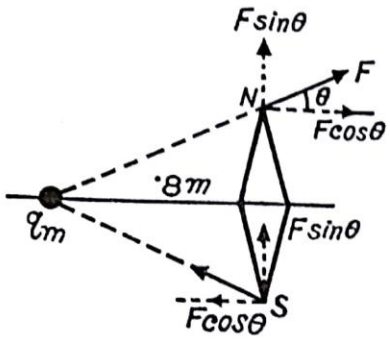


Fig. 1.3

$\therefore$ F (force on the N-pole of the needle)

$$= \frac{\mu_0}{4\pi} \times \frac{q_m \times 0.6}{0.8^2 + 0.06^2}.$$

The resolved part of the force perpendicular to the needle

$$= F\cos\theta = \frac{\mu_0}{4\pi} \cdot \frac{0.6 q_m}{0.8^2 + 0.06^2} \times \frac{0.8}{\sqrt{0.8^2 + 0.06^2}}$$

$$= 10^{-7} \times \frac{0.48\, q_m}{(0.6436)^{3/2}} = 0.93\ q_m \times 0.10^{-7}.$$

The same force is effective on the other pole.

$\therefore$ Torque on the needle = $(0.93 q_m \times 10^{-7}) \times 0.12 = 72 \times 10^{-7}$ (given)

$\therefore$ $$q_m = \frac{72}{0.93 \times 0.12} = 645 \text{ A m}.$$

19. At the neutral point on the broad-side-on position,

$$\frac{\mu_0}{4\pi} \cdot \frac{m}{x^3} = B_0 \text{ or } \frac{\mu_0}{4\pi} \cdot \frac{m}{15^3} = B_0.$$

When broken into two equal pieces, magnetic moment and length are halved.

$\therefore$ $$\frac{\mu_0}{4\pi} \cdot \frac{\frac{m}{2}}{x^3} = B_0 \qquad \therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{m}{2x^3} = B_0 = \frac{\mu_0}{4\pi} \cdot \frac{m}{0.15^3}$$

$$2x^3 = 0.15^3 \qquad \text{or} \qquad x = \frac{0.15}{2^{\frac{1}{3}}} = 0.15 \times 2^{-\frac{1}{3}} = 0.119 \text{ m}.$$

When the two pieces are placed together, the magnetic moment of the combination is $2\,\frac{m}{2}$.

$\therefore$ $$\frac{\mu_0}{4\pi} \cdot \frac{\frac{2m}{2}}{x^3} = B_0 = \frac{\mu_0}{4\pi} \cdot \frac{m}{0.15^3}$$

$\therefore$ $$x = 0.15 \text{ m}.$$

20. Let the weaker one make an angle θ with the magnetic meridian.

Then torque on (1) $\tau_1 = mB \sin\theta$

also torque on (2) $\tau_2 = 2mB \sin\left(\frac{\pi}{2} - \theta\right)$.

For equilibrium, $\tau_1 + \tau_2 = 0$ or $\tau_1 = -\tau_2$

$\therefore$ $\tau_1 = \tau_2$ (numerically)

or $mB \sin\theta = 2mB \sin\left(\frac{\pi}{2} - \theta\right) = 2mB \cos\theta$

or $\tan\theta = 2 = \tan 63°26'$ or $\theta = 63°26'$.

Hence combination will set with weaker magnet making an angle 63°26′ with magnetic meridian or with longer magnet making an angle 26°34′ with the magnetic meridian.

When the point is on the prolongation of m,

$$B_1 = \frac{\mu_0}{4\pi}\cdot\frac{2m}{d^3}; \quad B_2 = \frac{\mu_0}{4\pi}\cdot\frac{2m}{d^3}$$

$$B = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0 m}{4\pi d^3}\sqrt{2^2+2^2} = \frac{\mu_0}{4\pi}\cdot\frac{2\sqrt{2}m}{d^3}.$$

When the point is on the prolongation of $2m$,

$$B_1 = \frac{\mu_0}{4\pi}\cdot\frac{m}{d^3}; \quad B_2 = \frac{\mu_0}{4\pi}\cdot\frac{2(2m)}{d^3}.$$

$\therefore$ $$B = \frac{\mu_0}{4\pi}\cdot\frac{m}{d^3}\times\sqrt{1^2+4^2} = \frac{\mu_0}{4\pi}\cdot\frac{m\sqrt{17}}{d^3}.$$

21. We have $U = U_0 + mB(1 - \cos\theta)$

where U = energy in any position

U_0 = energy in the position when $\theta = 0$.

Assuming $U = 0$ when $\theta = \pi/2$ we have

$$0 = U_0 + mB\left(1 - \cos\frac{\pi}{2}\right) \quad \text{or} \quad U_0 = -mB.$$

$\therefore$ $$U = -mB + mB(1 - \cos\theta) = -mB\cos\theta = -\vec{m}\cdot\vec{B}.$$

In the position of stable equilibrium, potential energy is minimum.

From the above relation U (minimum) $= -mB$ when $\theta = 0°$.

22. Field at S' due to SN

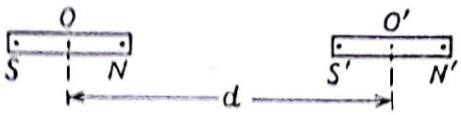


Fig. 3.4

$$= \frac{\mu_0}{4\pi} \cdot \frac{2m}{(d-l')^3}$$

$\therefore$ Force on S' due to SN

$$= \frac{\mu_0}{4\pi} \cdot \frac{2mq'_m}{(d-l')^3}.$$

Field at N' due to $SN = \frac{\mu_0}{4\pi} \cdot \frac{2m}{(d+l')^3}$.

$\therefore$ Force on N' due to $SN = \frac{\mu_0}{4\pi} \cdot \frac{2mq'_m}{(d+l')^3}$.

$\therefore$ Force by SN on $S'N'$

$$= \frac{\mu_0}{4\pi} \cdot \frac{2mq'_m}{(d-l')^3} - \frac{\mu_0}{4\pi} \cdot \frac{2mq'_m}{(d+l')^3}$$

$$= \frac{\mu_0}{4\pi} \cdot 2mq'_m \left[\frac{d^3 + 3d^2l' + 3dl'^2 + l'^3 - d^3 + 3d^2l' - 3dl'^2 + l'^{+3}}{(d-l')^3 (+l')^3} \right]$$

$$= \frac{\mu_0}{4\pi} \cdot 2mq'_m \frac{6d^2l' + 2l'^3}{(d^2 - l'^2)^3}.$$

$\therefore$ $d >> l'$,

$\therefore$ Mutual force $= \frac{\mu_0}{4\pi} \cdot 2mq'_m \frac{6d^2l'}{d^6} = \frac{\mu_0}{4\pi}.$

$$= \frac{6m(2q'_m l')}{d^4} = \frac{\mu_0}{4\pi} \cdot \frac{6mm'}{d^4}$$

23. Work done $= mB(1 - \cos 0) = \mu_0 mH(1 - \cos 0)$.

Here $0 = 90°$.

$\therefore$ Work done $= \mu_0 mH = 4\pi \times 10^{-7} \times 4 \times 30 = 1.5 \times 10^{-7}$ J.

24. The current loops act as dipoles of moments $m_1 = \pi r_1^2 I_1$ and $m_2 = \pi r_2^2 I_2$.

By the formula of interaction between dipoles (see Sol. of 22)

i.e., $F = \frac{\mu_0}{4\pi} \cdot \frac{6m_1m_2}{d^4}$,

$$F = \frac{\mu_0}{4\pi} \cdot \frac{6\pi^2 r_1^2 r_2^2 I_1 I_2}{d^4} = 10^{-7} \frac{6\pi^2 \times 1^2 \times 2^2 \times 10^{-4} \times 1 \times 2}{20^4 \times 10^{-8}}$$

$$= 2.3 \times 10^{-8} \text{ N}.$$

□ □ □

Chapter 4 – Magnetic Dipole

or $\tan\theta = 1.414$

or $\theta = 54°42'$

Also at the neutral point,

$$B_M = B_E$$

or $$\frac{\mu_0}{4\pi}\cdot\frac{m}{r^3}\sqrt{1+3\cos^2\theta} = 3.6\times10^{-5} \quad \text{or} \quad r^3 = \frac{10^{-7}\times2\sqrt{1+3\cos^2 54°42'}}{3.6\times10^{-5}}$$

or $$r^3 = \frac{\sqrt{1+3\times0.5778^2}\times10^{-2}}{1.8} = \frac{\sqrt{1+1.0017}\times10^{-2}}{1.8}$$

or $r = 0.199$ m.

Thus the neutral point will occur in a direction making 54°42′ with the axis and at a distance of 19.9 cm from the centre of the magnet.

3. *A magnet of magnetic moment* $3\ \text{A m}^2$ *when placed along the x-axis experiences a torque* $\vec{\tau} = 3\hat{j} + 4\hat{k}$ *and when placed along the y-axis, the torque* $\vec{\tau} = -3\hat{i} + \hat{j} - 6\hat{k}$ *find the magnetic field.*

Solution.

We have $\vec{\tau} = \hat{m}\times\hat{B}$

$$\therefore \quad 3\vec{j} + 4\vec{k} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & 0 \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}\cdot 0 + \hat{j}(-3B_z) + \hat{k}(3B_y)$$

$$\therefore \quad -3B_z = 3 \quad \text{or} \quad B_z = -1$$

and $\quad 3B_y = 4 \quad \text{or} \quad B_y = 4/3$

Again, $$-3\vec{i} + \vec{j} - 6\hat{k} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3 & 0 \\ B_x & 4/3 & -1 \end{vmatrix}$$

$$= -3\hat{i} + \hat{j} + \hat{k}(-3B_x)$$

$$\therefore \quad 6 = 3B_x \quad \text{or} \quad B_x = 2$$

$$\therefore \quad B = \sqrt{2^2 + 16/9 + 1} = 2.6 \text{ tesla}$$

The direction cosines of the field are $\frac{-3}{2.6}, \frac{1}{2.6}, \frac{2}{2.6}$

or $\quad -1.15, 0.38, 0.77$

EXERCISES

1. A very short magnet is placed at a point O with its axis horizontal and perpendicular to the magnetic meridian. If P is a neutral point, show that the angle between *OP* and the axis (SN direction) of the magnet is $\tan^{-1}\sqrt{2}$.
2. Find the magnitude and direction of the magnetic field due to a small bar magnet of moment 0.03 joule per tesla at a point situated on a line through the centre of the

magnet and at an angle of 60° with its axis, the point being at a distance of 0.05 m from the centre of the magnet.

3. Find the potential due to a magnet at a point on its axis at a distance of 0.09 m from its centre in the end-on position. The length of the magnet is 0.02 m and its pole strength 9 Am.

4. Two magnetic dipoles of moments 0.108 and 0.122 Am^2 are placed along two lines drawn on the table at right angles to each other. Find the intensity at the point of intersection of their axes, their centres being 0.3 and 0.4 m, respectively from the point of intersection.

5. Calculate the work done in transferring a unit magnetic pole from a point on the axis of a dipole of moment 0.3 joule per tesla, at a distance of 10 cm from its centre to a point on a line at an inclination 60° with the axis, at the same distance from the magnet.

6. Find the locus of the points where the intensity of the magnetic field due to a magnetic dipole will be perpendicular to its axis.

7. Two bar magnets are in line. One has a magnetic length of 0.08 m and pole strength of 5 A m and the other has a magnetic length of 0.04 m and pole strength of 4 A m, their midpoints being 0.16 m apart. Find the work done in turning one of these magnets over end for end.

8. The earth may be considered to be a big dipole of moment 6.4×10^{12} A m placed at its centre with its axis in the SN direction. The radius of the earth is 6000 km. Calculate the horizontal and vertical components of the earth's magnetic field at a place whose lattitude is 22° N.

[*Hint*: $B_r = \frac{\mu_0}{4\pi} \cdot \frac{2m \cos\theta}{r^3}$ and $B_\theta = \frac{\mu_0}{4\pi} \cdot \frac{m \sin\theta}{r^3}$; here $\theta = 90° + 22° = 112°$]

9. A point lies on a line making an angle of 30° with the axis (SN direction) of a magnetic dipole. If the dipole is turned through 90° away from the line, how will the magnetic field at the point be affected?

10. A pivoted small magnetic needle of moment m is placed at a distance r along the axis of a short bar magnet of moment M. Find the force on the pivot of the needle.

ANSWERS

2. 0.32×10^{-4} T at 100°54′ with the axis of the dipole
3. 2.23×10^{-6} $JA^{-1}m^{-1}$
4. 10^{-6} T
5. 0.15×10^{-5} J
6. At 54°42′ with the axis of the dipole
7. 7.4×10^{-6} J
8. $B_v = 2.22 \times 10^{-6}$ T (downward), $B_0 = 2.75 \times 10^{-6}$ T
9. Reduced by $\sqrt{\frac{7}{13}}$
10. $\frac{3\mu_0 mM}{2\pi r^4}$

CHAPTER 4

Magnetic Dipole

1. At the neutral point the earth's field is equal and opposite to the field due to the magnet.

$$\therefore \quad \theta + \alpha = \frac{\pi}{2}$$

but $\tan \alpha = \frac{\tan \theta}{2}$ (always)

$$\therefore \quad \tan\left(\frac{\pi}{2} - \theta\right) = \frac{\tan \theta}{2}$$

or $\cot \theta = \frac{\tan \theta}{2}$

$$\theta = \tan^{-1} \sqrt{2}.$$

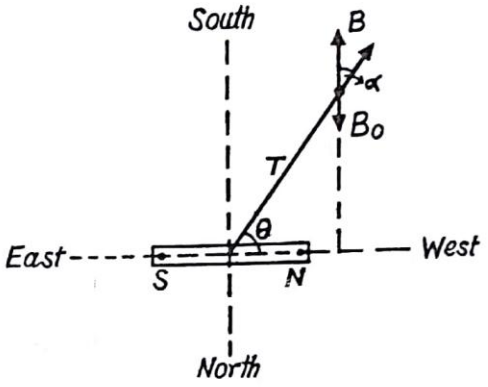


Fig. 4.1

2. $B = \frac{\mu_0}{4\pi} \cdot \frac{m}{d^3} \sqrt{1 + 3\cos^2 \theta}$

$$= 10^{-7} \cdot \frac{0.03}{0.5^3} \sqrt{1 + 3\cos^2 60°} = 0.32 \times 10^{-4} \text{ T}$$

$$\alpha = \tan^{-1} \frac{\tan 60°}{2} = \tan^{-1} 0.866 = 40°54'.$$

The angle made by the field with the axis = 60 + 40°54′ = 100° 54′.

3. $V_m(N)$ = magnetic potential at the point due to the north pole

$$= \frac{\mu_0}{4\pi} \cdot \frac{9}{0.08}$$

$V_m(S)$ = magnetic potential at the point due to the south pole

$$= \frac{\mu_0}{4\pi} \cdot \frac{-9}{0.09 + 0.01}$$

$$\therefore \quad V_m(NS) = \text{potential due to the magnet} = \frac{\mu_0}{4\pi} \cdot \left(\frac{9}{0.08} - \frac{9}{0.1}\right)$$

$$= 10^{-7} (112.5 - 90) = 2.25 \times 10^{-6} \text{ J A}^{-1} \text{ m}^{-1}.$$

4. $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2 \times 0.108}{0.3^3} = 10^{-7} \times \frac{2 \times 0.108}{0.027} = 8 \times 10^{-7} \text{ T}$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2 \times 0.192}{0.4^3} = 10^{-7} \times \frac{2 \times 0.192}{0.064} = 6 \times 10^{-7} \text{ T}$$

$$B = \sqrt{B_1^2 + B_2^2} = \sqrt{8^2 + 6^2} \times 10^{-7} = 10^{-6} \text{ tesla.}$$

5. V_m (magnetic potential at the first point)

$$= \frac{\mu_0}{4\pi} \times \frac{m \cos\theta}{r^2} = 10^{-7} \times \frac{0.3 \cos\theta}{0.1^2} = 3 \times 10^{-6}\ JA^{-1}\ m^{-1}$$

V'_m (magnetic potential at the second point) $= 10^{-7} \times \dfrac{0.3 \cos 60°}{0.1^2}$

$$= 1.5 \times 10^{-6}\ JA^{-1}\ m^{-1}$$

Work done = (potential difference) × magnetic pole

$= 1.5 \times 10^{-6} \times 1 = 1.5 \times 10^{-6}$ joule.

6. For any point we have

$$\theta + \alpha = 90° \quad \text{or} \quad \alpha = 90° - \theta$$

But $\tan\alpha = \dfrac{\tan\theta}{2}$

$\therefore$ $\tan(90 - \theta) = \dfrac{\tan\theta}{2}$ or $\tan\theta = \sqrt{2}$ or $\theta = 54°42'$.

Thus the locus of the points is a line at inclination 54°42′ with the axis of the dipole.

7. Potential at S' due to $SN = \dfrac{\mu_0}{4\pi} \times \left(\dfrac{5}{0.1} - \dfrac{5}{0.18} \right) = 2.222 \times 10^{-6}$

Potential at N' due to $SN = \dfrac{\mu_0}{4\pi} \times \left(\dfrac{5}{0.14} - \dfrac{5}{0.22} \right) = 1.300 \times 10^{-6}$

$\therefore$ Potential energy of $S'N'$ due to SN

$$= 4 \times (2.222 - 1.300) \times 10^{-6} = 3.688 \times 10^{-6}.$$

When reversed end for end the potential energy $= -3.688 \times 10^{-6}$

$\therefore$ Work done to turn over end for end

$$= 3.688 \times 10^{-6} - (-3.688 \times 10^{-6})$$

$$= 7.376 \times 10^{-6} = 7.4 \times 10^{-6} \text{ joule.}$$

8. We have

B_r (radial component of the field) $= \dfrac{\mu_0}{4\pi} \times \dfrac{2m \cos\theta}{r^3} = B_v$

(vertical component)

B_p (transverse component of the field) $= \dfrac{\mu_0}{4\pi} \times \dfrac{m \sin\theta}{r^3} = B_0$

(horizontal component)

where θ is the angle from the axis ($\overrightarrow{SN}$) in the anticlockwise direction and r is the distance of the point from the centre of the magnet.

Here $\theta = 90° + 22°$ and r = radius of the earth = 6000 km

$= 6 \times 10^6$ m.

$\therefore \quad B_v = 10^{-7} \times \dfrac{2 \times 6.4 \times 10^{21} \times \cos\ (90° + 22°)}{(6 \times 10^6)^3} = -2.2 \times 10^{-6}$ T.

$$B_0 = 10^{-7} \times \frac{6.4 \times 10^{21} \sin (90° + 22°)}{(6 \times 10^6)^3} = 2.75 \times 10^{-6} \text{ T.}$$

9. We have $B = \dfrac{\mu_0}{4\pi} \times \dfrac{m}{r^3} \times \sqrt{1 + 3\cos^2\theta} = \dfrac{\mu_0}{4\pi} \cdot \dfrac{m}{r^3}\ \sqrt{1 + 3\cos^2 30°}$

$$= \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3} \cdot \frac{\sqrt{13}}{2}.$$

When turned through 90°, θ becomes $\theta + 90°$ or $\theta + 270°$.

In either case we have $\cos^2 (\theta + 90°) = \sin^2 \theta$

$\cos^2 (\theta + 270°) = \sin^2 \theta$.

$$\therefore \quad B' = \frac{\mu_0}{4\pi} \times \frac{m}{r^3}\ \sqrt{1 + 3\sin^2 30°} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3} \cdot \frac{\sqrt{7}}{2}$$

$$\therefore \quad \frac{B}{B'} = \sqrt{\frac{13}{7}} \quad \text{or} \quad B' = \sqrt{\frac{7}{13}} \times B.$$

Thus the field is decreased $\sqrt{\dfrac{7}{13}}$ times.

10. We have $B = \dfrac{\mu_0}{4\pi} \times \dfrac{2M}{r^3}$ and $F = m \dfrac{\partial B}{\partial l}$.

$$\therefore \quad F = m \frac{\mu_0}{4\pi} \times 2M \frac{\partial}{\partial r} \left(\frac{1}{r^3} \right)$$

$$= m \frac{\mu_0}{4\pi} \times 2M\ (-3)\ r^{-4} = -\frac{3\mu_0 mM}{2\pi r^4}.$$

The negative sign shows that there is force of attraction.

$$\therefore \quad \text{Force} = \frac{3\mu_0 mM}{2\pi r^4} \text{ (attractive).}$$

□ □ □

EXERCISES
(A)

1. Two short bar magnets are placed one above the other with their (a) similar poles, (b) dissimilar poles together. The first combination is placed at a distance of 0.2 m from the needle of a deflection magnetometer in the tan A position. The deflection is observed to be 60°. The second combination is placed at half the distance in the tan B position and the deflection is now 30°. Compare the moments of the magnets.
2. A bar magent of length 0.1 m, breadth 0.02 m and thickness 0.005 m is placed at a distance of 0.3 m from the centre of a deflection magnetometer arranged in the tan A position and it produces a deflection of 38°. Calculate the magnetization of the magnet. (Horizontal induction of earth $= 3 \times 10^{-5}$ tesla.)
3. A short bar magnet is placed on the arm of a deflection magnetometer arranged in the usual manner in the tan B position and it produces a deflection of 60°. It is now broken into two equal parts and one part is placed exactly at the same distance. What deflection does it produce? At what distance should it be placed to produce the same deflection as before? If the other piece is placed on the top of it at right angles to it, will there be any change in deflection?
4. A compass needle vibrates 10 times per minute in the earth's field alone. When the north pole of a long magnet is brought towards the south of the needle and 10 cm away from it, it makes 20 vibrations per minute. Calculate the number of vibrations of the needle when the distance of the pole of the magnet is reduced to 8 cm.
5. A rectangular bar magnet of length 0.1 m and breadth 0.01 m oscillates once in 10 s in the earth's horizontal field of intensity 18×10^{-6} T. If the mass of the magnet is 40×10^{-3} kg, find the magnetic moment of the magnet.
6. Two short bar magnets exactly alike, of magnetic moment 60×10^{-3} A m^2 and 80×10^{-3} A m^2, are fastened together with their axes parallel and horizontal and their centres in a vertical line about which the system is free to oscillate. Compare the time of oscillation in the earth's field when the axes of magnets are in the same direction with that of when they are in the opposite direction.
7. A small magnet vibrating in the earth's horizontal field completes 4 oscillations in 16 seconds and when another small magnet is brought near it, it oscillates 5 times in 16 seconds. Compare the intensity of the second magnet with the earth's horizontal intensity when (i) both the fields are in the same direction (ii) in opposite directions.
8. An iron cylinder 1 m in length and 0.02 m in diameter is uniformly magnetized. Then 0.05 m is cut from it and placed on the frame of an oscillation magnetometer. It makes 10 oscillations in 30 seconds. Calculate magnetization of the rod. (Density of iron $= 7000$ kg m^{-3} and $B = 4 \times 10^{-5}$ tesla.)
9. A thin iron rod which is 30 cm long is uniformly magnetized and its time of oscillation is 4s. It is broken into 3 equal parts. Find the time of oscillation of each part.
10. Two bar magnets, the moment of one of which is double that of the other, but otherwise similar, are tied together so as to form a cross. The cross is now freely suspended in such a way that it may oscillate in the horizontal plane. How will cross rest in the earth's field? When it is set to oscillations, what will its time period be if the time period of the weaker magnet in a similar position is 5 sec?
11. A compass needle makes 10 oscillations per minute in the earth's field alone. A bar

magnet is placed in such a way that its field at the position of the needle is at right angles to the earth's field and it deflects the needle through 45°. If the needle is made to oscillate in the deflected position, find the number of oscillations per minute.

12. Two identical bar magnets are tied together at their centres at an inclination of 60° with each other. Calculate the time period of oscillations of the combination when the time period of each magnet is 2 s.

13. A magnet of moment 0.5 Am^2 is deflected through an angle of 60° in a magnetic field of strength 4×10^{-5} tesla and then released. Its angular velocity at the instant it passes through its equilibrium position is 0.5 radian per second. Calculate its moment of inertia.

14. A short magnet produces a deflection of 45° in a deflection magnetometer when placed in the usual manner. A similar magnet is placed on it with like poles together. What deflection do the two magnets jointly produce?

15. A bar magnet measuring 3 cm × 2 cm × 1 cm is placed on the frame of an oscillation magnetometer with its 2-cm edge vertical. Its time period of oscillation in the earth(s) field is 5 s. What will be the time period when (a) the 1-cm edge is vertical, 3-cm edge is vertical?

16. A short bar magnet when suspended horizontally and perpendicular to the earth's field experiences a torque of 3×10^{-4} Nm. The same magnet produces a deflection of 45° in the needle of a deflection magnetometer set in the tan A position when the distance of the magnet from the centre of the magnetometer is 0.3 m. Calculate the moment of the magnet.

17. A bar magnet weighing 250 g and measuring 10 cm × 2 cm × 1 cm is suspended by unspun silk fibre passing through its centre of gravity with the 1-cm edge vertical. It takes 1 minute 40 s to make 20 oscillations. Calculate mB_0 of the magnet. Also calculate the torque required to deflect it through 90° in the earth's field and the work done by the torque.

(B)

18. P is a neutral point on the perpendicular bisector of a bar magnet. A small compass needle is placed at P. The bar magnet is then removed and the needle is set oscillating. The bar magnet is then carefully replaced in its position, but with its north pole pointing south and the needle is again set oscillating. Compare the periodic times of oscillation in the two cases.

19. A deflection magnetometer is arranged with its arms along the east-west direction. A small bar magnet with its north pole pointing east is placed lengthwise along the eastern arm. The needle is found to be deflected through 30°. What would be the deflection of the needle if the magnetometer arms were rotated anticlockwise through (a) 90°, (b) 45°?

20. A bar magnet of length 10 cm and breadth 2 cm weighs 200 g. When it is placed on the frame of an oscillation magnetometer, it makes 20 oscillations in 1 minute 20 seconds. It produces a deflection of 45° when the distance of its nearer pole from the needle of a deflection magnetometer arranged in the tan A position is 10 cm. Calculate magnetic moment and B-vector of the earth's magnetic field along the horizontal.

[*Hint:* $mB_0 = \frac{4\pi^2 I}{t^2}$ and $\frac{m}{B_0} = \frac{4\pi}{\mu_0} \cdot \frac{(d^2 - l^2)^2}{2d} \tan\theta$]

21. A deflection magnetometer is set in the tan *A* position. A short bar magnet is placed at a certain distance on the arms of the magnetometer with its north pole pointing north. There is no deflection. When the needle of the magnetometer is set to oscillation by an auxiliary magnet, it makes 5 oscillations in one minute. When the magnet is turned through 180° in its own position, the needle makes 20 oscillations per minute. What permanent deflection will be produced when the magnet will be along the arm of the magnetometer at the same position?

ANSWERS

1. 13 : 11 **2.** 2.99×10^5 A m^{-1}

3. 40°54′; 0.793 times the previous distance; yes, restoring force will be changed

4. 23.8 vibrations/minute **5.** 0.74 Am^2 **6.** 1 : 2.646 **7.** (i) 9 : 16, (ii) 41 : 16

8. 1.792×10^5 Am^{-1} **9.** 1.333 s **10.** 63°30′; 4.73 s **11.** 11.87 per minute

12. 2.15 s **13.** 8×10^{-5} kg m^2 **14.** 63°26′ **15.** 5.7 s; 3.54 s **16.** 6.37 Am^2

17. 3.42×10^{-4} J, 3.42×10^{-7} Nm, 3.42×10^{-4} J **18.** 1.414 : 1

19. (a) no deflection; (b) 16° **20.** 2.39 A m^2, 1.8×10^{-4} T **21.** 60°24′

CHAPTER 5

Magnetic Measurements

(A)

1. Let m_1 and m_2 be the magnetic moments of individual magnets. Then the magnetic moments of the combination will be $(m_1 + m_2)$ when similar poles are together and $(m_1 - m_2)$ when dissimilar poles are together.

$$\therefore \quad \frac{\mu_0}{4\pi} \times \frac{2(m_1 + m_2)}{d^3} = B_0 \tan 60° \qquad \ldots \text{(i)}$$

where B_0 = horizontal component of B-vector of the earth's field

and $$\frac{\mu_0}{4\pi} \times \frac{m_1 - m_2}{\left(\frac{d}{2}\right)^3} = B_0 \tan 30°. \qquad \ldots \text{(ii)}$$

Dividing, $$\frac{2(m_1 + m_2)}{8(m_1 - m_2)} = \frac{\tan 60°}{\tan 30°} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} = 3$$

or $$m_1 + m_2 = 12\, m_1 - 12\, m_2 \quad \text{or} \quad \frac{m_1}{m_2} = \frac{13}{11}.$$

2. We have in tan–A position $\frac{\mu_0}{4\pi} \times \frac{2md}{(d^2 - l^2)^2} = B_0 \tan \theta$.

$$\therefore \quad 10^{-7} \frac{2m \times 0.3}{(0.3^2 - 0.05^2)^2} = 3 \times 10^{-5} \times \tan 38°$$

or $$m = 5 \times 875^2 \times 0.7813 \times 10^{-6}$$

volume of the magnet = $0.1 \times 0.02 \times 0.005 = 10^{-5}\ \text{m}^3$

$\therefore$ M, magnetization of the magnet

$$= \frac{5 \times 875^2 \times 0.7813 \times 10^{-6}}{10^{-5}}$$

$$= 2.99 \times 10^5\ \text{A m}^{-1}.$$

3. We have in tan–B position, $\frac{\mu_0}{4\pi} \times \frac{m}{d^3} = B_0 \tan 60°$. ... (i)

When broken into two equal parts, the magnetic moment is halved.

$$\therefore \quad \frac{\mu_0}{4\pi} \cdot \frac{\frac{m}{2}}{d^3} = B_0 \tan 0. \qquad \ldots \text{(ii)}$$

Dividing, $\tan\theta = \dfrac{\tan 60°}{2'} = \dfrac{1.732}{2} = 0.866$

or $\theta = 40°54'$.

Let d' be the distance at which broken piece will produce 60° deflection.

Then $\dfrac{\mu_0}{4\pi} \times \dfrac{m}{2d'^3} = B_0 \tan 60°,$ ... (iii)

From (i) and (iii),

$$d^3 = 2d'^3$$

or $$d' = \frac{d}{2^{\frac{1}{3}}} = 2^{-\frac{1}{3}} \times d = 0.793d.$$

4. We have $10^2 = KB_0$. ... (i)

Since the magnet is long, the effect of the distant pole on the needle may be neglected.

$\therefore$ $20^2 = K(B_0 + B_1)$ ... (ii)

where B_1 = magnetic field due to the pole.

$$n^2 = K(B_0 + B_2),$$ ... (iii)

From (i) and (ii),

$$\frac{400}{100} = \frac{B_0 + B_1}{B_0} \quad \text{or} \quad B_1 = 3B_0$$

and from (i) and (iii),

$$\frac{n^2}{100} = \frac{B_0 + B_2}{B_0} \quad \text{or} \quad B_2 = \frac{n^2 - 100}{100} \times B_0$$

$\therefore$ $$\frac{B_1}{B_2} = \frac{300}{n^2 - 100}.$$

Since due to a pole,

$$B \propto \frac{1}{d^2}$$

$\therefore$ $$\frac{B_1}{B_2} = \frac{8^2}{10^3} = \frac{64}{100}$$

$\therefore$ $$\frac{64}{100} = \frac{300}{n^2 - 100}$$

or $n = 23.8$.

5. We have $t = 2\pi \sqrt{\dfrac{I}{mB}}$

where I = moment of inertia, m = magnetic moment of the magnet

and B = intensity of the magnetic field.

Here $I = w \times \frac{a^2 + b^2}{12} = 40 \times 10^{-3} \left(\frac{0.1^2 + 0.01^2}{12} \right) = 336 \times 10^{-7}$ km^2

$\therefore \quad m = \frac{4\pi^2 I}{t^2 B}$

or $m = \frac{4\pi^2 \times 336 \times 10^{-7}}{10^2 \times 18 \times 10^{-6}} = 0.74$ A m^2.

6. We have $t_1 = 2\pi \sqrt{\frac{I_1 + I_2}{(60 + 80) \times 10^{-3} B_0}}$

$$t_2 = 2\pi \sqrt{\frac{I_1 + I_2}{(80 - 60) \times 10^{-3} \times B_0}}$$

$\therefore \quad \frac{t_1}{t_2} = \sqrt{\frac{20}{140}} = \sqrt{\frac{1}{7}} = \frac{1}{2.646}$.

7. We have $K \left(\frac{4}{16} \right)^2 = B_0$ ... (i)

and $K \left(\frac{5}{16} \right)^2 = (B_0 + B)$ ... (ii)

Dividing $\frac{25}{16} = \frac{B_0 + B}{B_0}$ or $\frac{B}{B_0} = \frac{9}{16}$.

When the two fields are in opposite directions the resultant field is either $B - B_0$ or $B_0 - B$

Hence either $B - B_0 = K \left(\frac{5}{16} \right)^2$ or $B_0 - B = K \left(\frac{5}{16} \right)^2$.

Dividing,

$$\frac{B - B_0}{B_0} = \frac{25}{16} \quad \text{or} \quad \frac{B}{B_0} = \frac{41}{16}.$$

The other case will be found to be absurd in the same way.

8. Mass = $\pi (0.01)^2 \times 0.05 \times 7000 = 35\pi \times 10^{-3}$ kg.

$$I = 35\pi \times 10^{-3} \left(\frac{0.05^2}{12} + \frac{0.01^2}{4} \right) = 2.566 \times 10^{-5} \text{ kg m}^2.$$

We have $t = 2\pi \sqrt{\frac{I}{mB'}}$, $\quad 3 = 2\pi \sqrt{\frac{2.566 \times 10^{-5}}{m \times 4 \times 10^{-5}}}$

or $9 = \frac{4\pi^2 \times 0.6415}{m}$ or $m = \frac{4\pi^2 \times 0.6415}{9}$

M (magnetization) $= \frac{m \text{ (magnetic moment)}}{V \text{ (volume)}} = \frac{4\pi^2 \times 0.6415}{9 \times \pi (0.01)^2 \times 0.05}$

$= 1.792 \times 10^5$ A m^{-1}.

9. Since the rod is thin, its radius is negligible in comparison to its length.

$\therefore \quad I = 30w \times \frac{30^2}{12}$ where w is the mass per unit length

or $\quad I = \frac{27000w}{12}$

m (magnetic moment) $= q_m \times 30 = 30\, q_m$

where q_m is the pole strength of the rod.

$$\therefore \quad 4 = 2\pi \sqrt{\frac{\frac{27000}{12}w}{30\, q_m B_0}} = 2\pi \sqrt{\frac{900w}{12 q_m B_0}}. \qquad \ldots \text{(i)}$$

When broken into three equal parts, mass is reduced three times, length is reduced three times but pole strength remains same.

$$I' = 10w.\frac{10^2}{12} = \frac{1000}{12}w \quad \text{and} \quad m' = q_m \times 10 = 10q_m$$

$$\therefore \quad t = 2\pi \sqrt{\frac{\frac{1000\,w}{12}}{10 q_m B_0}} = 2\pi \sqrt{\frac{100w}{12 q_m B_0}}. \qquad \ldots \text{(ii)}$$

Dividing (ii) by (i), $\frac{t}{4} = \sqrt{\frac{1}{9}} = \frac{1}{3}$ or $t = 1.333$ s.

10. Let the weaker one make an angle θ with the earth's field.

Then $\quad \tau_1 = mB_0 \sin\theta$ and $\tau_2 = 2mB_0 \sin(90 - \theta)$.

For equilibrium, $mB_0 \sin\theta = 2mB_0 \sin(90 - \theta)$

or $\quad \tan\theta = 2$ or $\theta = \tan^{-1} 2 = 63°30'$.

The combination will behave like a single magnet of moment

$$m' = \sqrt{m^2 + (2m)^2} = m\sqrt{5}$$

and moment of inertia, $I' = 2I$ where I is the moment of inertia of either magnet.

Now, $\quad 5 = 2\pi \sqrt{\frac{I}{mB_0}}$ and $t = 2\pi \sqrt{\frac{2I}{\sqrt{5}\, mB_0}}$

$$\therefore \quad \frac{t}{5} = \sqrt{\frac{2}{\sqrt{5}}} \quad \text{or} \quad t = 5 \sqrt{\frac{2}{2.236}} = 4.73 \text{ s}.$$

11. We have $\quad 10^2 = KB_0 \qquad \ldots \text{(i)}$

$$B = B_0 \tan\theta = B_0 \tan 45° = B_0.$$

Resultant field $= \sqrt{B^2 + B_0^2} = \sqrt{B_0^2 + B_0^2} = B_0\sqrt{2}$.

$$\therefore \quad n^2 = KB_0\sqrt{2}. \qquad \ldots \text{(ii)}$$

Dividing, $\frac{n^2}{10^2} = \sqrt{2}$ or $n = \sqrt{141.4} = 11.87$.

12. The combination will behave like a single magnet of moment

$$m' = \sqrt{\sqrt{m^2 + m^2 + 2m.m \cos 60°} = m\sqrt{3}.}$$

We have $2 = 2\pi\sqrt{\frac{I}{mB_0}}$ and $t = 2\pi\sqrt{\frac{2I}{\sqrt{3}\,mB_0}}$

$\therefore$ $\frac{t}{2} = \sqrt{\frac{2}{\sqrt{3}}}$ or $t = 2.15$ s.

13. Energy in the deflected position

$$= U_0 + 0.5 \times 4 \times 10^{-5}(1 - \cos 60°)$$

Energy in the equilibrium position

$$= U_0 + \frac{1}{2} I\omega^2 = U_0 + \frac{1}{2} I (0.05)^2.$$

By the principle of conservation of energy,

$$U_0 + \frac{1}{2} I (0.5)^2 = U_0 + 2 \times 10^{-5}(1 - \cos 60°)$$

or $\frac{1}{2} I (0.5)^2 = 2 \times 10^{-5} \times \frac{1}{2}$

or $I = 8 \times 10^{-5}$ kg m^2.

14. Let the magnetometer be in tan-B position.

Then $\frac{\mu_0}{4\pi} \cdot \frac{m}{d^3} = B_0 \tan 45°$.

When the second magnet is placed over the first with like poles together the magnetic moment of the combination is 2 m.

$\therefore$ $\frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3} = B_0 \tan \theta$.

Hence $\theta = 63°26'$.

15. **When 2** cm edge is vertical, $I = w \times \left(\frac{0.03^2 + 0.01^2}{12} \right) = \frac{w}{12} \times 10^{-3}$.

$$\therefore \quad 5 = 2\pi \sqrt{\frac{\frac{w}{12} \times 10^{-3}}{mB_0}} \cdot \qquad \ldots \text{(i)}$$

When 1 cm edge is vertical,

$$I = w \left(\frac{0.03^2 + 0.02^2}{12} \right) = \frac{w}{12} \times 1.3 \times 10^{-3}$$

$$\therefore \quad t = 2\pi \sqrt{\frac{\frac{w}{12} \times 1.3 \times 10^{-3}}{mB_0}} \cdot \qquad \ldots \text{(ii)}$$

Dividing (ii) by (i),

$$\frac{t}{5} = \sqrt{1.3} \qquad \text{or} \quad t = 5\sqrt{1.3} = 5.7 \text{ s.}$$

When 3 cm edge is vertical,

$$I = w\left(\frac{0.02^2 + 0.01^2}{12}\right) = \frac{w}{12} \times 0.5 \times 10^{-3}$$

$$t = 2\pi\sqrt{\frac{\frac{w}{12} \times 0.5 \times 10^{-3}}{mB_0}}. \qquad \ldots \text{(iii)}$$

Dividing (iii) by (i),

$$\frac{t}{5} = \sqrt{0.5} \qquad \text{or} \qquad t = 5\sqrt{0.5} = 3.54 \text{ s.}$$

16. We have $\tau = mB_0 \sin\theta$

$\therefore$ $3 \times 10^{-4} = mB_0 \sin 90° = mB_0$. ... (i)

In tan-A position we have for short magnets,

$$\frac{\mu_0}{4\pi} \times \frac{2m}{d^3} = B_0 \tan\theta$$

or $$\frac{\mu_0}{4\pi} \times \frac{2m}{(0.3)^3} = B_0 \tan 45°. \qquad \ldots \text{(ii)}$$

Dividing (i) by (ii),

$$m = 6.37 \text{ A m}^2.$$

17. $I = 250 \times 10^{-3}\left(\frac{0.1^2 + 0.02^2}{12}\right) = 2.166 \times 10^{-4} \text{ kg m}^2.$

We have for oscillation magnetometer,

$$t = 2\pi\sqrt{\frac{I}{mB_0}} \quad \text{or} \quad mB_0 = \frac{4\pi^2 I}{t^2}$$

$$mB_0 = \frac{4\pi^2 \times 2.166 \times 10^{-4}}{(5)^2} = 3.42 \times 10^{-4} \text{ joule.}$$

$$\tau = mB_0 \sin\theta = 3.42 \times 10^{-4} \sin 90° = 3.42 \times 10^{-4} \text{ N m.}$$

Work done $= mB_0(1 - \cos\theta) = mB_0(1 - \cos 90°)$

$= 3.42 \times 10^{-4}$ joule.

(B)

18. We have at the neutral point $B = B_0$

where B = magnetic field due to the magnet on the broad-side-on position and B_0 = magnetic B-field due to earth's field. When the magnet is removed, we have only one field and that field is B_0.

$$t_1^2 = \frac{K}{B_0} \quad \left(\because \text{ in an oscillation magnetometer } t^2 \propto \frac{1}{B}\right).$$

When the magnet is replaced, the total field at the same point is $B + B_0$ or $2B_0$ because now the two fields are in the same direction.

$$\therefore \qquad t_2^2 = \frac{K}{2B_0}$$

$$\therefore \qquad \frac{t_1^2}{t_2^2} = \frac{2}{1} \quad \text{or} \quad t_1 : t_2 = 1.414 : 1.$$

19. We have in tan-A position, $B = B_0 \tan 30°$

$$\text{or} \qquad \frac{\mu_0}{4\pi} \cdot \frac{2m}{d} = B_0 \tan 30°. \qquad \ldots (i)$$

(a) When rotated through 90°, B and B_0 are in the same direction. Hence there is no deflection.

(b) When rotated through 45°, the angle between B_0 and B is 45°. If the needle makes an angle θ with B_0 then

$$\tau_1 = mB_0 \sin\theta \quad \text{and} \quad \tau_2 = mB \sin(45° - \theta)$$

where m is the magnetic moment of the needle.

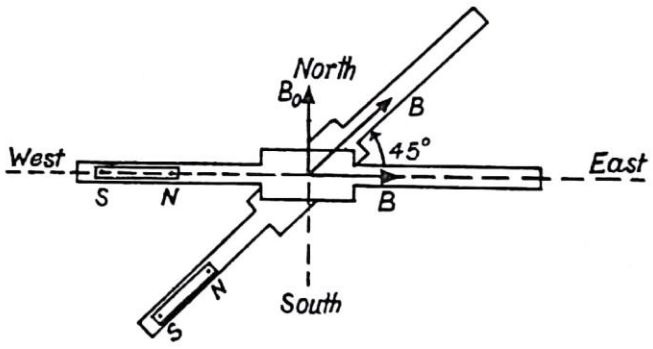


Fig. 5.1

For equilibrium,

$$mB_0 \sin\theta = mB \sin(45° - \theta)$$

$$\text{or} \qquad B_0 \sin\theta = B \sin(45° - \theta)$$

$$\text{or} \qquad B_0 \sin\theta = \frac{\mu_0}{4\pi} \cdot \frac{2m}{d^3} \sin(45° - \theta) \qquad \ldots (ii)$$

where m is the magntic moment of the magnet.

Dividing (ii) by (i), $\sin(45 - \theta) = \dfrac{\sin\theta}{\tan 30°} = \sqrt{3}\sin\theta$

$$\text{or} \qquad \frac{1}{\sqrt{2}} \cdot \cos\theta - \frac{1}{\sqrt{2}} \sin\theta = \sqrt{3}\sin\theta$$

$$\text{or} \qquad \cos\theta - \sin\theta = \sqrt{6}\sin\theta \quad \text{or} \quad \cos\theta = (\sqrt{6} + 1)\sin\theta$$

$$\text{or} \qquad \tan\theta = \frac{1}{\sqrt{6}+1} = 0.29 \quad \text{or} \quad \theta = 16°.$$

20. I (moment of inertia of the magnet) $= 200 \times 10^{-3}\left(\frac{0.1^2 + 0.02^2}{12}\right)$

$= 17.3 \times 10^{-5}$ kg m^2.

We have for oscillation magnetometer,

$$t = 2\pi\sqrt{\frac{I}{mB_0}}$$

or $$mB_0 = \frac{4\pi^2 I}{t^2} = \frac{4\pi^2 \times 17.3 \times 10^{-5}}{4^2} = 42.7 \times 10^{-5}. \quad \ldots \text{(i)}$$

In tan-A position we have

$$\frac{m}{B_0} = \frac{4\pi}{\mu_0} \cdot \frac{(d^2 - l^2)^2}{2d} \tan\theta$$

or $$\frac{m}{B_0} = 10^7 \times \frac{(0.15^2 - 0.05^2)^2}{2 \times 0.15} \times \tan 45° \qquad (\because \mu_0 = 4\pi \times 10^{-7})$$

or $$\frac{m}{B_0} = 1.333 \times 10^4. \quad \ldots \text{(ii)}$$

Multiplying (i) and (ii),

$$m^2 = 42.7 \times 1.333 \times 10^{-1} = 5.7$$

$$m = 2.39 \text{ A m}^2.$$

Dividing (i) by (ii),

$$B_0^2 = \frac{42.7 \times 10^{-4}}{1.333 \times 10^4} = 3.2 \times 10^{-8},$$

or $$B_0 = 1.8 \times 10^{-4} \text{ tesla}.$$

21. The two fields are along the same line. This is why there is no deflection of the needle. The resultant field at the site of the needle is $B_0 - B$.

$\therefore$ $$B_0 - B = K \times 5^2. \quad \ldots \text{(i)}$$

When the magnet is turned through 180°, B and B_0 are in the same direction.

$\therefore$ $$B + B_0 = K \times 20^2.$$

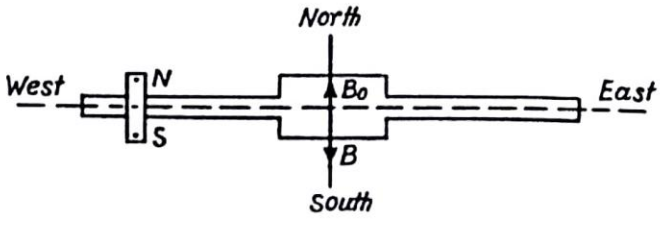


Fig. 5.2

$$\therefore \qquad \frac{B + B_0}{B_0 - B} = \frac{400}{25} = 16 \qquad \text{or} \qquad B = \frac{15}{17} B_0 = \frac{\mu_0}{4\pi} \times \frac{m}{d^3}.$$

When the magnet is set along the arm, B' and B_0 are at right angles to each other and now $B' = \frac{\mu_0}{4\pi} \times \frac{2m}{d^3}$.

$$\therefore \qquad B' = B_0 \tan \theta$$

$$\text{or} \qquad \frac{\mu_0}{4\pi} \times \frac{2m}{d^3} = B_0 \tan \theta \qquad \text{or} \qquad \tan \theta = \frac{2 \times 15}{17} = 1.76$$

$$\theta = 60°24'.$$

□ □ □

$$\therefore \quad \frac{25}{16} = \frac{B_0}{B_0'}. \quad \text{Also } B_0 = B\cos 30° \text{ and } B_0' = B'\cos 60°$$

$$\therefore \quad \frac{B_0}{B_0'} = \frac{B\cos 30°}{B'\cos 60°} \qquad \therefore \quad \frac{25}{16} = \frac{B}{B'}\cdot\frac{\cos 30°}{\cos 60°}$$

or $$\frac{B}{B'} = \frac{25}{16}\times\frac{\cos 60°}{\cos 30°} = \frac{25}{16}\times\frac{0.5}{0.866}$$

or $$B : B' = 0.902 : 1$$

Similarly, the H-vectors of the fields at the two places

$$H : H' = 0.902 : 1$$

EXERCISES

1. Calculate the total induction at a place where the horizontal component is 25×10^{-6} T, and the dip 45°.
2. The vertical component of the earth's field at a place is 40 A m^{-1}. Calculate the value of H_0, if dip of the place is 30°.
3. The apparent dip at a place in a plane 30° away from the magnetic meridian is 60°. Calculate the true dip at the place.
4. The apparent dips in two mutually perpendicular planes are found to be δ_1 and δ_2. Show that true dip δ is related with δ_1 and δ_2 by

$$\cot^2\delta = \cot^2\delta_1 + \cot^2\delta_2$$

[*Note:* This is the principle of cot-method of finding true dip of a place.]

5. Considering the earth's magnetism to be due to a very powerful short magnet embedded at its centre placed with its south pole pointing north, show that, latitude (λ) of any place bears a definite relation with the dip (δ) of the place and that definite relation is $\tan\delta = 2\tan\lambda$.
6. The earth is a big dipole of moment 64×10^{22} A m^2. Calculate the horizontal and vertical components of the earth's magnetic field at a place of latitude 30° South. (Radius of the earth = 6000 km)

[*Hint:* See example 5.]

7. A dip needle indicates a dip of 60° at a place. When a small magnet is placed on the horizontal line through the centre of the dip needle with its north pole pointing north at a distance 0.2 m from the needle, the dip changes to 45°. Find the magnetic moment of the magnet if the horizontal component of the earth's field is 0.2×10^{-4} tesla.
8. The magnetic moment of a dip needle is 0.5 A m^2 and it is set at a place where dip is 60°. A weight of 0.05 g placed 4 cm from the axle causes the needle to turn the horizontal position. Calculate the value of the horizontal and vertical components of the earth's field.
9. The period of oscillation of a dip needle when vibrating in the magnetic meridian is 1.5 s. In a plane at right angles to the magnetic meridian it is 2s. Find the dip of the place.
10. In an experiment of finding dip by the cot-method it is found that the apparent dips

in two mutually perpendicular planes are 30° and 20°. Calculate the true dip of the place.

11. The needle of a dip circle completes 4 oscillations in one minute while vibrating in a vertical plane at right angles to the megnetic meridian. The same needle oscillates 3 times per minute in a horizontal plane at the same place. Find the value of dip at the place.

12. A magnetic needle is suspended by a thread at its centre and it shows a dip of 60°. When a weight of 20 mg is placed at its front end, the dip is reduced to 30°. If the vertical component of the earth's field is 4×10^{-5} tesla, find the pole strength of the needle.

13. Calculate the earth's total induction (B-vector) at a place where the *H*-vector of the magnetic field along the horizontal is 30 Am^{-1} and the dip 40°.

[***Hint:*** $H_0 = H \cos \delta \quad \therefore \quad H = \dfrac{H_0}{\cos \delta}$; $B = \mu_0 H$.]

14. A magnet vibrating horizontally at a place where the angle of dip is 45° and the total intensity of the earth's field is 40 A m^{-1}, makes 10 oscillations per minute. Calculate the number of oscillations that it would make per minute at another place where the dip is 60° and the total intensity 50 A m^{-1}.

15. A dip circle is set in position after levelling and the dip is found to be α. Then it is rotated through 45° away from the magnetic meridian from this position and the dip is found to be β. Show that true dip is given by

$$\cot^2 \delta = 2(\cot^2 \alpha + \cot^2 \beta - \sqrt{2} \cot \alpha \cot \beta)$$

ANSWERS

1. 35.35×10^{-6} T **2.** 69.28 Am^{-1} **3.** 56°18′ **6.** 3×10^{-5} T, 2.5×10^{-5} T

7. 0.585 Am^2 **8.** $B_v = 0.392 \times 10^{-4}$ T, $B_0 = 0{:}225 \times 10^{-4}$ T **9.** 34°15′ **10.** 17°7′

11. 60°36′ **12.** $3.6\dot{7}$ Am **13.** 4.9×10^{-5} T **14.** 9.4 per minute

CHAPTER 6

Terrestrial Magnetism

1. We have $B_0 = B\cos\delta \quad \therefore \quad 25\times10^{-6} = B\cos 45°$

$B = 25\sqrt{2}\times10^{-6} = 35.35\times10^{-6}$ T.

2. $\tan\delta = \dfrac{H_v}{H_0} \quad \therefore \quad \tan 30° = \dfrac{40}{H_0}$

or $H_0 = \dfrac{40}{\tan 30°} = 40\sqrt{3} = 69.28 \text{ A m}^{-1}.$

3. We have $\tan\delta' = \dfrac{\tan\delta}{\cos\alpha} \quad \therefore \quad \tan 60° = \dfrac{\tan\delta}{\cos 30°}$

or $\delta = 56°18'.$

4. We have $\tan\delta' = \dfrac{\tan\delta}{\cos\alpha}.$

$\therefore \quad \tan\delta_1 = \dfrac{\tan\delta}{\cos\alpha}$ and $\tan\delta_2 = \dfrac{\tan\delta}{\cos(90-\alpha)}$

or $\cos\alpha = \dfrac{\tan\delta}{\tan\delta_1}$ and $\sin\alpha = \dfrac{\tan\delta}{\tan\delta_2}.$

Squaring and adding,

$$1 = \cos^2\alpha + \sin^2\alpha = \tan^2\delta\left(\frac{1}{\tan^2\delta_1} + \frac{1}{\tan^2\delta_2}\right)$$

or $\cot^2\delta = \cot^2\delta_1 + \cot^2\delta_2.$

5. See figure 6 · 4 of the textbook.

$$B_0 = \frac{\mu_0}{4\pi}\times\frac{m\sin\theta}{r^3}; \quad B_v = \frac{\mu_0}{4\pi}\times\frac{2m\cos\theta}{r^3}.$$

Here $\theta = 90° + \lambda$

$$B_0 = \frac{\mu_0}{4\pi}\times\frac{m\sin(90°+\lambda)}{r^3} = \frac{\mu_0}{4\pi}\times\frac{m\cos\lambda}{r^3}$$

and $$B_v = \frac{\mu_0}{4\pi}\times\frac{2m\cos(90+\lambda)}{r^3} = -\frac{\mu_0}{4\pi}\times\frac{2m\sin\lambda}{r^3}.$$

The minus sign shows that the verticle component is towards the centre of the earth.

Now, $\tan\delta = \dfrac{B_v}{B_0} \quad \therefore \quad \tan\delta = \dfrac{2\sin\lambda}{\cos\lambda} = 2\tan\lambda.$

6. We have $B_0 = \frac{\mu_0}{4\pi} \times \frac{m \cos \lambda}{r^3} = 10^{-7} \times \frac{6.4 \times 10^{22} \cos 30°}{(6 \times 10^6)^3}$

$= 2.5 \times 10^{-5}$ T.

$B_v = \frac{\mu_0}{4\pi} \cdot \frac{2m \sin \lambda}{r^3} = 10^{-7} \times \frac{2 \times 6.4 \times 10^{22} \times \sin 30°}{(6 \times 10^6)^3}$

$= 3 \times 10^{-5}$ T.

7. We have $\tan 60° = \frac{B_v}{B_0}$ or $B_v = \sqrt{3}\, B_0$.

When the magnet is placed in the position given in the question, the total horizontal induction is the sum of the two fields.

$\therefore$ The total horizontal induction $= B_0 + B$

where B is the induction of the magndetic field on the end-on position of the magnet.

$\therefore$ $\tan 45° = \frac{B_v}{B_0 + B}$ or $B_0 + B = B_v = \sqrt{3}\, B_0$

or $B = (1.732 - 1)\, B_0 = 0.732 B_0$

or $\frac{\mu_0}{4\pi} \times \frac{2m}{0.2^3} = 0.732\, B_0$ or $10^{-7} \times \frac{2m}{0.008} = 0.732 \times 0.2 \times 10^{-4}$

$m = \frac{0.732 \times 0.2 \times 0.008 \times 10^{-4}}{2 \times 10^{-7}} = 0.585$ A m^2.

8. We have $\tan 60° = \frac{B_v}{B_0}$ or $\sqrt{3}\, B_0 = B_v$

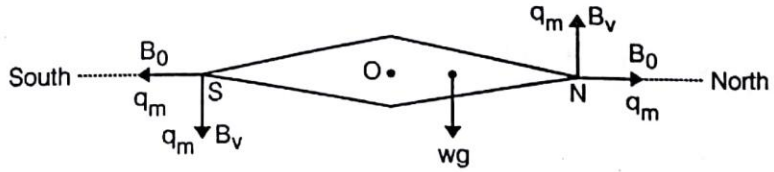


Fig. 6.1

Taking moment about the axle in the horizontal position we have

$q_m B_v \times 2l - 0.05 \times 10^{-3} \times 9.8 \times 4 \times 10^{-2} = 0$

or $B_v m = 2 \times 10^{-6} \times 9.8$ $(\because \quad m = 2q_m l)$

or $B_v = \frac{2 \times 10^{-6} \times 9.8}{0.5} = 4 \times 10^{-6} \times 9.8 = 0.392 \times 10^{-4}$ T.

$\therefore$ $B_0 = \frac{B_v}{\sqrt{3}} = \frac{0.392}{\sqrt{3}} \times 10^{-4} = 0.225 \times 10^{-4}$ T.

9. In the magnetic meridian the needle oscillates in the total field.

$$\therefore \qquad B = k\left(\frac{1}{1.5}\right)^2 = \frac{k}{2.25} \qquad \left[\because\ B \propto \left(\frac{1}{T}\right)^2\right]$$

or $$\sqrt{B_0^2 + B_v^2} = \frac{k}{2.25}$$

or $$B_0^2 + B_v^2 = \frac{k^2}{(2.25)^2}. \qquad \ldots \text{(i)}$$

In the plane at right angles to the magnetic meridian the dip needle is under the influence of the vertical component only.

$$\therefore \qquad B_v = k\left(\frac{1}{2}\right)^2 = \frac{k}{4} \quad \text{or} \quad B_v^2 = \frac{k^2}{16}. \qquad \ldots \text{(ii)}$$

Dividing

$$\frac{B_0^2 + B_v^2}{B_v^2} = \frac{k^2}{(2.25)^2} \times \frac{16}{k^2} = \frac{16}{(2.25)^2} = \frac{16}{5}$$

or $$\frac{B_0}{B_v} = \sqrt{\frac{11}{5}}.$$

Now, $$\tan\delta = \frac{B_v}{B_0} = \sqrt{\frac{5}{11}} = 0.6742$$

$$\delta = 34°15'.$$

10. We have $$\cot^2\delta = \cot^2\delta_1 + \cot^2\delta_2$$

$$= \cot^2 30° + \cot^2 20°$$

$$= 3 + 7.548 = 10.548$$

$$\therefore \qquad \cot\delta = \sqrt{10.548} = 3.2478 \quad \text{or} \quad \delta = 17°7'.$$

11. In the first case the dip needle is under the influence of vertical component of the earth's field only.

$$\therefore \qquad B_v = k \times 4^2.$$

In the horizontal plane it is effected by the horizontal component only.

$$\therefore \qquad B_0 = k \times 3^2$$

$$\therefore \qquad \frac{B_v}{B_0} = \frac{16}{9} \quad \text{or} \quad \tan\delta = \frac{16}{9} = 1.778$$

or $$\delta = 60°36'.$$

12.

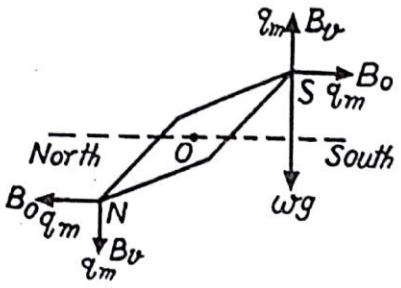


Fig. 6.2

We have $\tan 60° = \dfrac{B_v}{B_0}$

or $\sqrt{3}\, B_0 = B_v = 4 \times 10^{-5}$

or $B_0 = 2.31 \times 10^{-5}$

Taking moment about the centre

$q_m B_v 2l \cos 30°$

$- q_m B_0 2l \sin 30°$

$-20 \times 10^{-6} \times 9.8\, l \cos 30° = 0$

where q_m is the pole strength and l is half the length of the needle

or $2q_m \times \sqrt{3}\, B_0 \dfrac{\sqrt{3}}{2} - 2q_m B_0 \times \dfrac{1}{2} - 20 \times 10^{-6} \times 9.8 \dfrac{\sqrt{3}}{2} = 0$

or $q_m\, 3 \times 2.3 \times 10^{-5} - q_m \times 2.31 \times 10^{-5} - 17 \times 10^{-5} = 0$

or $q_m = \dfrac{17}{4.62} = 3.67$ A m.

13. We have $H_0 = H \cos \delta$ or $H = \dfrac{H_0}{\cos \delta} = \dfrac{30}{\cos 40} = \dfrac{30}{0.766} = 39.16.$

Now $B = \mu_0 H = 4\pi \times 10^{-7} \times 39.16 = 4.9 \times 10^{-5}$ T.

14. We have $B_0 = \mu_0 H_0 = \mu_0 H \cos \delta = \mu_0\, 40 \cdot \cos 45°$

and $\mu_0\, 40 \cdot \cos 45° = K \cdot 10^2$

or $20 \sqrt{2}\, \mu_0 = K \cdot 100.$ ... (i)

At the second place,

$\mu_0 \times 50 \cos 60° = Kn^2$

or $\mu_0 25 = Kn^2$ ... (ii)

Dividing (ii) by (i),

$$\frac{n^2}{100} = \frac{\mu_0 25}{\mu_0 20\, \sqrt{2}} = \frac{5}{4\, \sqrt{2}}$$

or $n^2 = \dfrac{125}{\sqrt{2}}$

or $n = 9.4$ oscillations per min.

15. Let the plane of the dip circle be $\theta°$ away from the magnetic meridian.

Then $\tan \alpha = \dfrac{\tan \delta}{\cos \theta}$ or $\cos \theta = \dfrac{\tan \delta}{\tan \alpha}$ ... (i)

When rotated through 45° further the plane is now $(\theta + 45°)$ away from the magnetic meridian.

$$\therefore \quad \tan\beta = \frac{\tan\delta}{\cos(45+\theta)} \quad \text{or} \quad \cos(45+\theta) = \frac{\tan\delta}{\tan\beta}$$

$$\text{or} \quad \frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta = \frac{\tan\delta}{\tan\beta}$$

$$\text{or} \quad \frac{\tan\delta}{\tan\alpha} - \sin\theta = \frac{\sqrt{2}\tan\delta}{\tan\beta}$$

$$\text{or} \quad \sin\theta = \tan\delta\cot\alpha - \sqrt{2}\tan\delta\cot\beta$$

$$\text{or} \quad \sin\theta = \tan\delta(\cot\alpha - \sqrt{2}\cot\beta) \qquad \ldots \text{(ii)}$$

Squaring and adding (i) and (ii),

$$\cot^2\delta = 2(\cot^2\alpha + \cot^2\beta - \sqrt{2}\cot\alpha \,.\, \cot\beta).$$

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Appendix

Conversion Factors

Length:
$1\ m = 10^2\ cm = 39.37\ in$
$= 6.214 \times 10^{-4}\ mi$
$1\ mi = 5280\ ft = 1.609\ km$
$1\ in = 2.540\ cm$
$1\ Å\ (angstrom) = 10^{-8}\ cm$
$= 10^{-10}\ m$
$= 10^{-4}\ \mu\ (micron)$
$1\ \mu\ (micron) = 10^{-6}\ m$
$1\ AU\ (astronomical\ unit)$
$= 1.496 \times 10^{11}\ m$
$1\ light\ year = 9.46 \times 10^{15}\ m$
$1\ parsec = 3.084 \times 10^{16}\ m$

Angle:
$1\ radian = 57.3°$
$1° = 1.74 \times 10^{-2}\ rad$
$1' = 2.91 \times 10^{-4}\ rad$
$1'' = 4.85 \times 10^{-6}\ rad$

Area:
$1\ m^2 = 10^4\ cm^2 = 1.55 \times 10^{-5}\ in^2$
$= 10.76\ ft^2$
$1\ in^2 = 6.452\ cm^2$
$1\ ft^2 = 144\ in^2 = 9.29 \times 10^{-2}\ m^2$

Volume:
$1\ m^3 = 10^6\ cm^3 = 10^3\ litres$
$= 35.3\ ft^3 = 6.1 \times 10^4\ in^3$
$1\ ft^3 = 2.83 \times 10^{-2}\ m^3$
$= 28.32\ litres$
$1\ in^3 = 16.39\ cm^3$

Mass:
$1\ kg = 10^3\ g = 2.205\ lb$
$1\ lb = 453.6\ g = 0.4536\ kg$
$1\ amu = 1.6604 \times 10^{-27}\ kg$

Force:
$1\ N = 10^5\ dyn + 0.2248\ lbf$
$= 0.102\ kgf$
$1\ dyn = 10^{-5}\ N$
$= 2.248 \times 10^{-6}\ lbf$
$1\ lbf = 4.448\ N$
$= 4.448 \times 10^5\ dyn$
$1\ kgf = 9.81\ N$

Power:
$1\ W = 1.341 \times 10^{-3}\ hp$
$1\ hp = 745.7\ W$

Electric Charge:*
$1\ C = 3 \times 10^9\ stC$*
$1\ stC = \frac{1}{3} \times 10^{-9}\ C$

Current:*
$1\ A = 3 \times 10^9\ stA$
$1\ stA = \frac{1}{3} \times 10^{-9}\ A$
$1\ \mu A = 10^{-6}\ A,\ 1\ mA = 10^{-3}\ A$

Electric Field:*
$1\ N\ C^{-1} = 1\ V\ m^{-1}$
$= 10^{-2}\ V\ cm^{-1}$
$= \frac{1}{3} \times 10^{-4}\ stV\ cm^{-1}$

Electric Potential:*
$1\ V = \frac{1}{3} \times 10^{-2}\ stV$*
$1\ stV = 3 \times 10^2\ V$

Resistance:
$1\ \Omega = 10^6\ \mu\Omega$
$1\ M\Omega = 10^6\ \Omega$

Capacitance:*
$1\ F = 9 \times 10^{11}\ stF$*
$1\ stF = \frac{1}{9} \times 10^{-11}\ F$
$1\ \mu F = 10^{-6}\ F,\ 1\ pF = 10^{-12}\ F$

Magnetic Field:
$1\ T = 10^4\ gauss$
$1\ gauss = 10^{-4}\ T$

Magnetic Flux:
$1\ Wb = 10^8\ maxwell$
$1\ maxwell = 10^{-8}\ Wb$

Magnetizing Field:
$1\ A\ m^{-1} = 4\pi \times 10^{-3}\ oersted$
$1\ oersted = 1/4\pi \times 10^3\ A\ m^{-1}$

* In all cases 3 actually means 2.998 and 9 means 8.987; stC—stat coulomb (esu of charge); stV—stat volt (esu of voltage).

Some Physical Constants

Speed of light	c	3×10^{8} m s^{-1}
Gravitational constant	G	6.67×10^{-11} N m^{2} kg^{-2}
Universal gas constant	R	8.31 J mol^{-1} K^{-1}
Permeability constant	μ_0	$4\pi \times 10^{-7}$ H m^{-1}
Permittivity constant	ε_0	8.85×10^{-12} F m^{-1}
Avogadro constant	N	6.02×10^{23} mol^{-1}
Boltzmann constant	k	1.38×10^{-23} J K^{-1}
Planck constant	h	6.63×10^{-34} J s
Elementary charge	e	1.60×10^{-19} C
Electron rest mass	m_e	9.11×10^{-31} kg
Electron magnetic moment	μ_e	9.27×10^{-24} J T^{-1}
Acceleration due to gravity (standard)	g	9.80 m s^{-2}
Standard atmospheric pressure	atm	1.01×10^{5} N m^{-2}
Density of water at 4°C		1000 kg m^{-3}
Rydberg constant	R	1.097×10^{7} m^{-1}
Loschmidt constant	n	2.69×10^{25} m^{-3}
Specific charge of electron	e/m	1.76×10^{11} C kg^{-1}
Proton rest mass	m_p	1.672×10^{-27} kg
Proton specific charge	e/m_p	9.59×10^{8} C kg^{-1}
Stefan constant	σ	5.67×10^{-8} W m^{-3} K^{-4}
Wien constant	b	2.9×10^{-3} m V
First Bohr orbit	r_0	5.29×10^{-11} m
Classical electron radius	r_e	2.82×10^{-15} m
Bohr magneton	μ_B	9.27×10^{-24} J T^{-1}
Nuclear magneton	μ_N	5.05×10^{-27} J T^{-1}
Proton magnetic moment	μ_p	2.792 μN
Neutron magnetic moment	μ_n	− 1.913 μN
Atomic mass unit	amu	1.66×10^{-27} kg or 931 MeV
	hc	1.989×10^{-25} J m
	$\frac{hc}{e}$	1.243×10^{-6} V m

□